HCLTECH CLOUD DATA ENGINEERING

ASSESSMENT 2

SUBMITTED BY

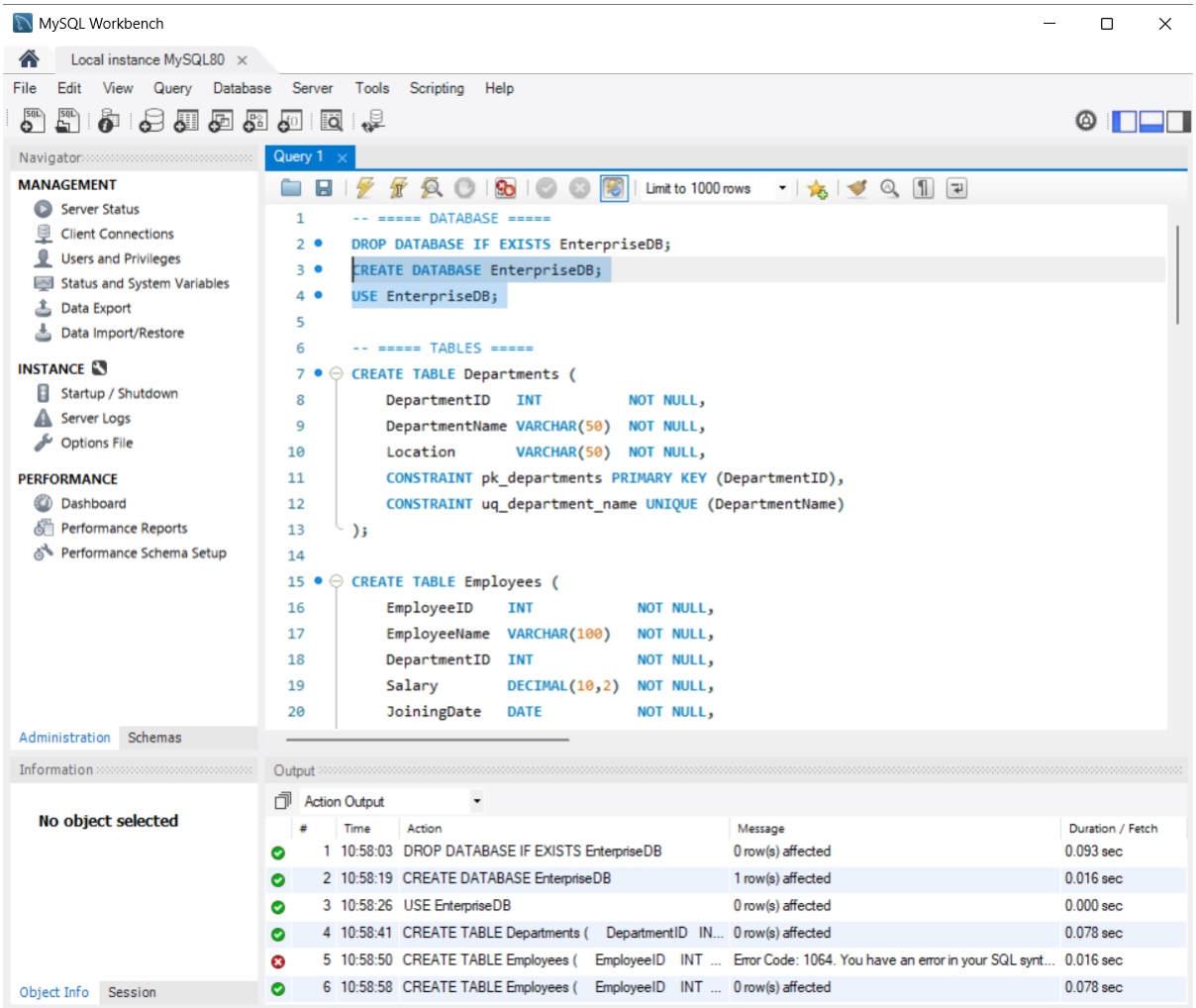
Kaviya

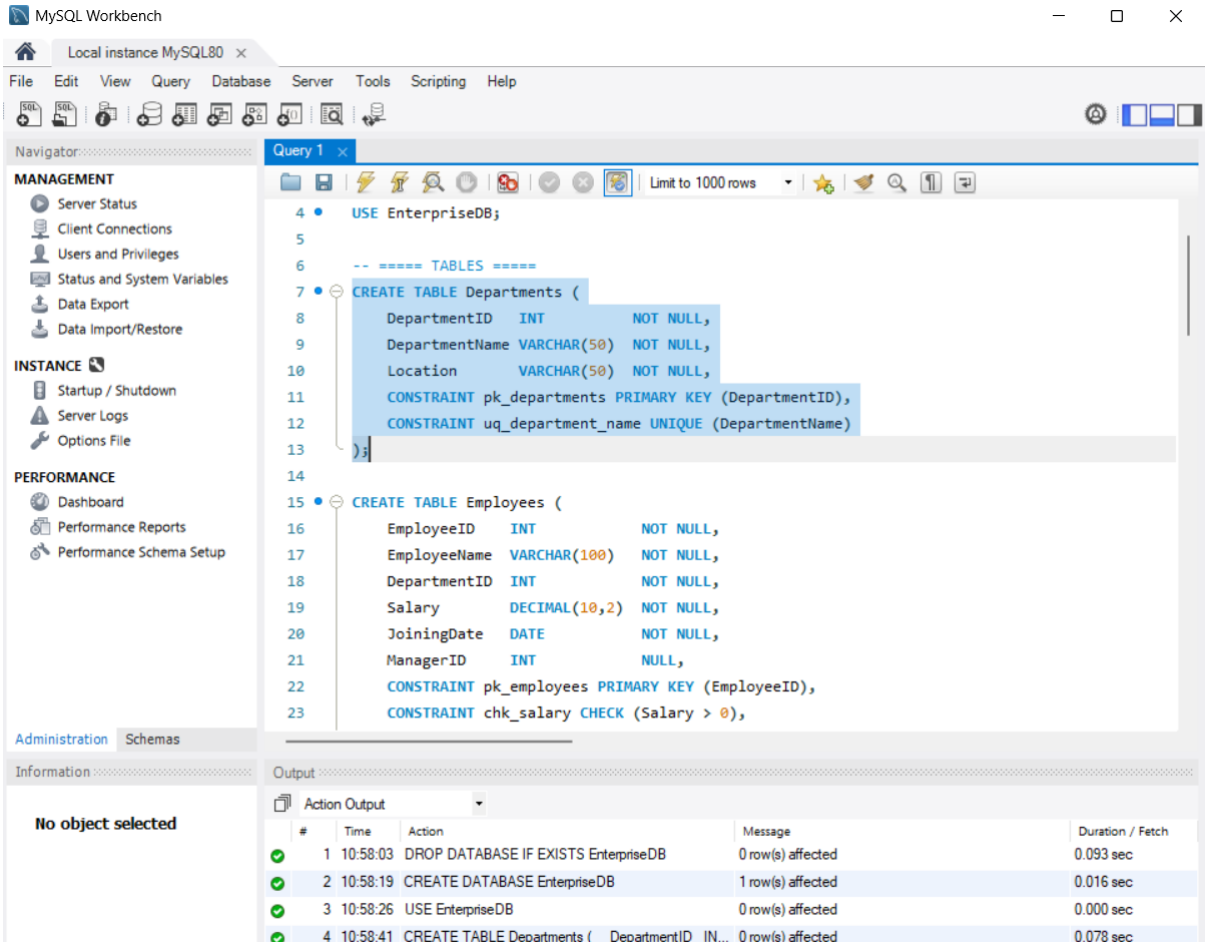
USN: 23btrcl049

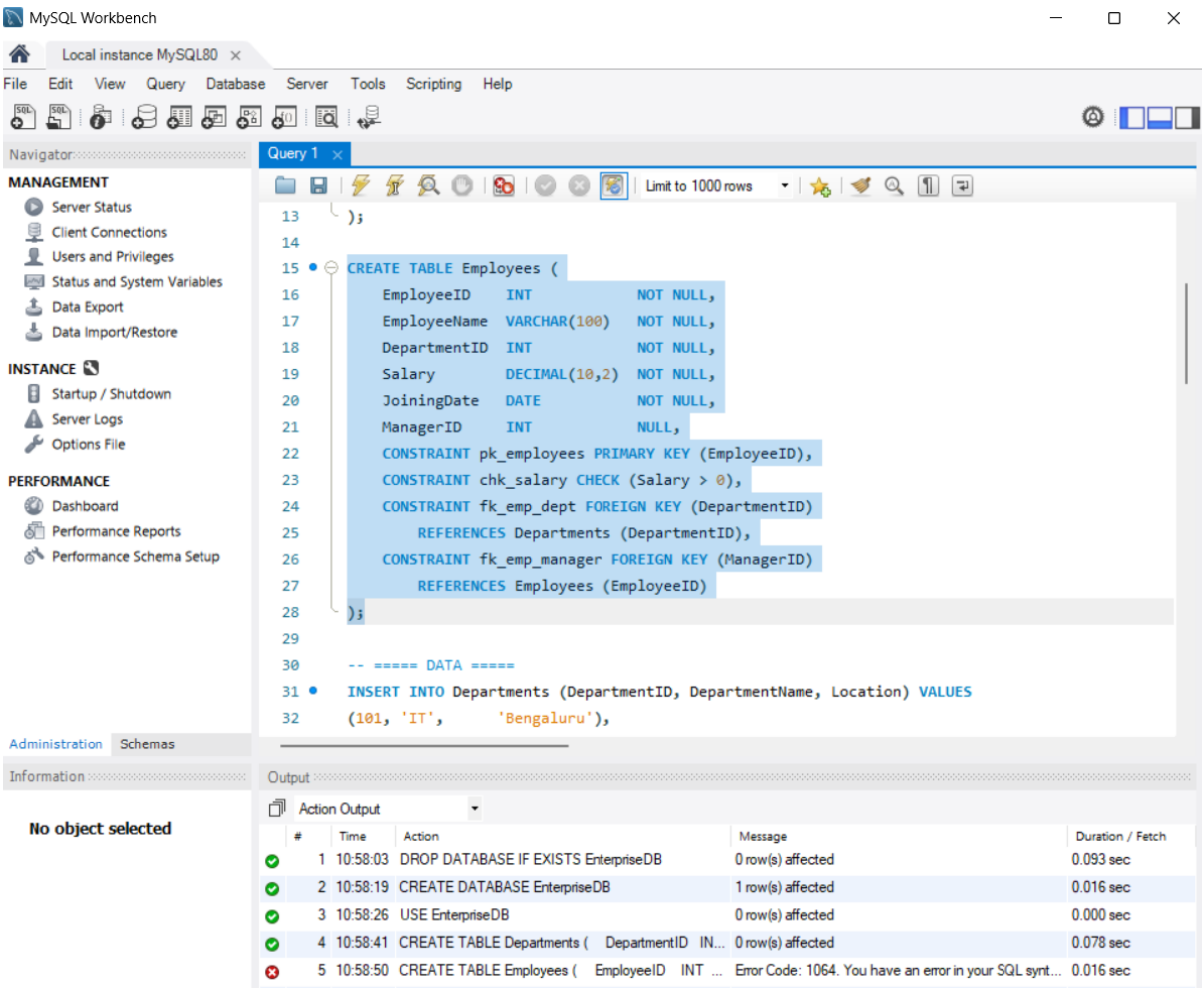
Jain University

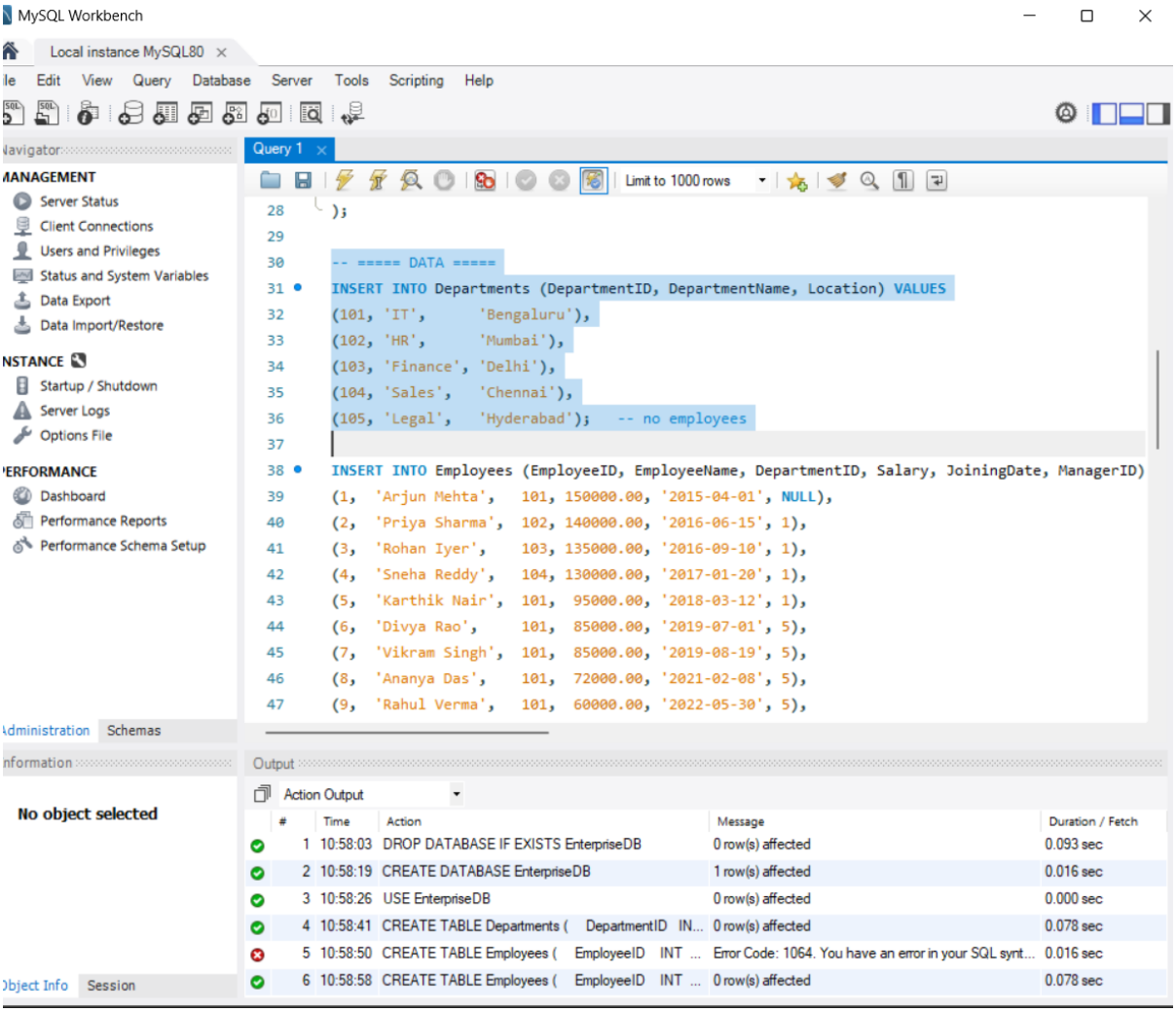
assengment 2
Question 1

Question 1: Database Design and Advanced Queries (8 Marks)
Create a database named EnterpriseDB. Create the Employees table (EmployeeID, EmployeeName, DepartmentID, Salary, JoiningDate, ManagerID) and the Departments table (DepartmentID, DepartmentName, Location), with appropriate data types and constraints, primary keys and an appropriate foreign key. Insert at least 15 employee records and five department records, including employees with different salaries, departments and managers, and at least one department without employees.

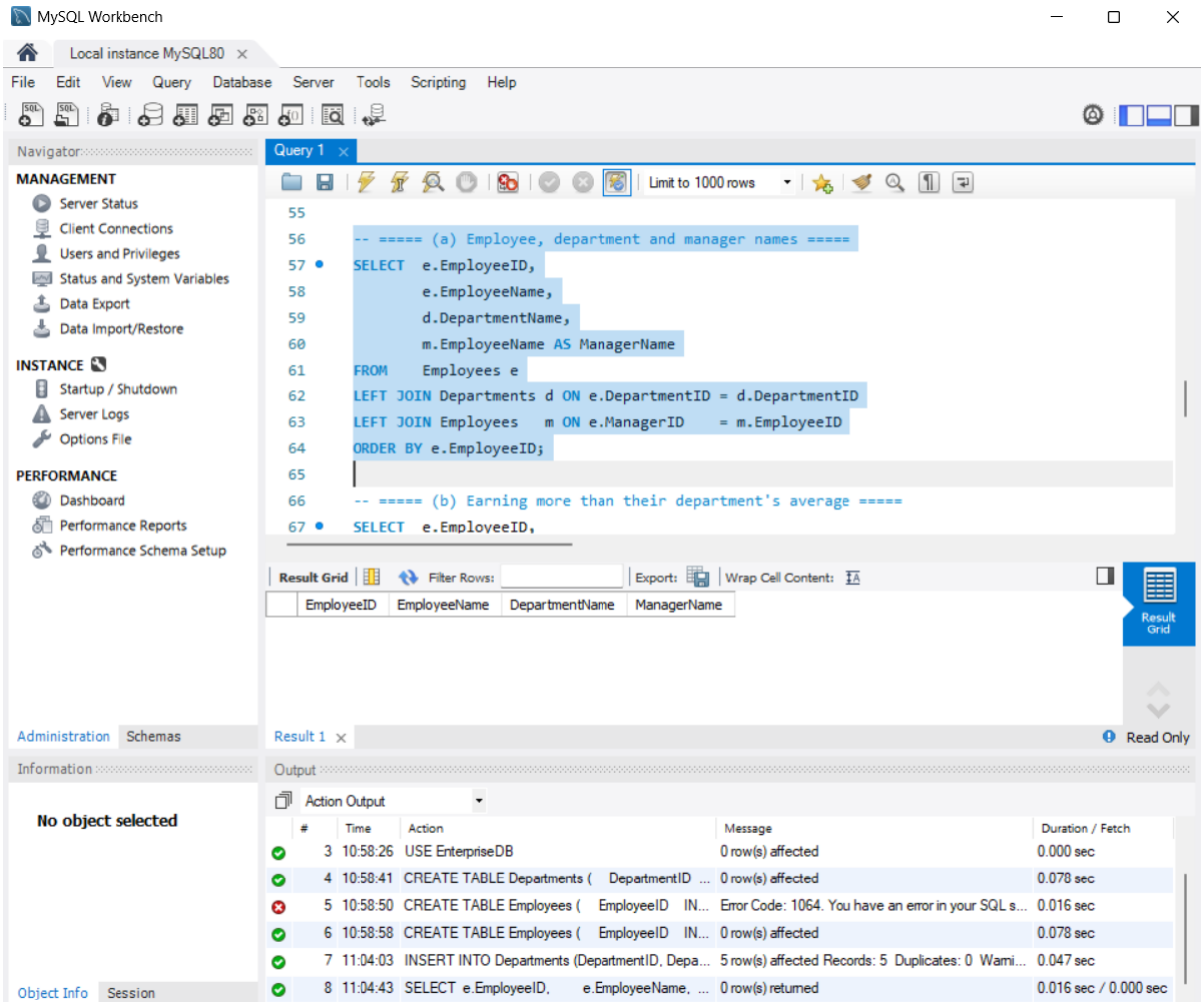

MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
MANAGEMENT
Server Status
Client Connections
Users and Privileges
Status and System Variables
Data Export
Data Import/Restore
INSTANCE
Startup / Shutdown
Server Logs
Options File
PERFORMANCE
Dashboard
Performance Reports
Performance Schema Setup
Administration Schemas
Information
No object selected
Object Info Session
Query 1
Limit to 1000 rows
1 -- ===== DATABASE =====
2 DROP DATABASE IF EXISTS EnterpriseDB;
3 CREATE DATABASE EnterpriseDB;
4 USE EnterpriseDB;
5
6 -- ===== TABLES =====
7 CREATE TABLE Departments (
8 DepartmentID INT NOT NULL,
9 DepartmentName VARCHAR(50) NOT NULL,
10 Location VARCHAR(50) NOT NULL,
11 CONSTRAINT pk_departments PRIMARY KEY (DepartmentID),
12 CONSTRAINT uq_department_name UNIQUE (DepartmentName)
13);
14
15 CREATE TABLE Employees (
16 EmployeeID INT NOT NULL,
17 EmployeeName VARCHAR(100) NOT NULL,
18 DepartmentID INT NOT NULL,
19 Salary DECIMAL(10,2) NOT NULL,
20 JoiningDate DATE NOT NULL,
Output
Action Output
Time Action Message Duration / Fetch
1 10:58:03 DROP DATABASE IF EXISTS EnterpriseDB 0 row(s) affected 0.093 sec
2 10:58:19 CREATE DATABASE EnterpriseDB 1 row(s) affected 0.016 sec
3 10:58:26 USE EnterpriseDB 0 row(s) affected 0.000 sec
4 10:58:41 CREATE TABLE Departments (DepartmentID IN... 0 row(s) affected 0.078 sec
5 10:58:50 CREATE TABLE Employees (EmployeeID INT ... Error Code: 1064. You have an error in your SQL synt... 0.016 sec
6 10:58:58 CREATE TABLE Employees (EmployeeID INT ... 0 row(s) affected 0.078 sec


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
MANAGEMENT
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Status and System Variables
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INSTANCE
Startup / Shutdown
Server Logs
Options File
PERFORMANCE
Dashboard
Performance Reports
Performance Schema Setup
Administration Schemas
Information
No object selected
Query 1
Limit to 1000 rows
4 USE EnterpriseDB;
5
6 -- ===== TABLES =====
7 CREATE TABLE Departments (
8 DepartmentID INT NOT NULL,
9 DepartmentName VARCHAR(50) NOT NULL,
10 Location VARCHAR(50) NOT NULL,
11 CONSTRAINT pk_departments PRIMARY KEY (DepartmentID),
12 CONSTRAINT uq_department_name UNIQUE (DepartmentName)
13);
14
15 CREATE TABLE Employees (
16 EmployeeID INT NOT NULL,
17 EmployeeName VARCHAR(100) NOT NULL,
18 DepartmentID INT NOT NULL,
19 Salary DECIMAL(10,2) NOT NULL,
20 JoiningDate DATE NOT NULL,
21 ManagerID INT NULL,
22 CONSTRAINT pk_employees PRIMARY KEY (EmployeeID),
23 CONSTRAINT chk_salary CHECK (Salary > 0),
Output
Action Output
Time Action Message Duration / Fetch
1 10:58:03 DROP DATABASE IF EXISTS EnterpriseDB 0 row(s) affected 0.093 sec
2 10:58:19 CREATE DATABASE EnterpriseDB 1 row(s) affected 0.016 sec
3 10:58:26 USE EnterpriseDB 0 row(s) affected 0.000 sec
4 10:58:41 CREATE TABLE Departments (DepartmentID IN... 0 row(s) affected 0.078 sec

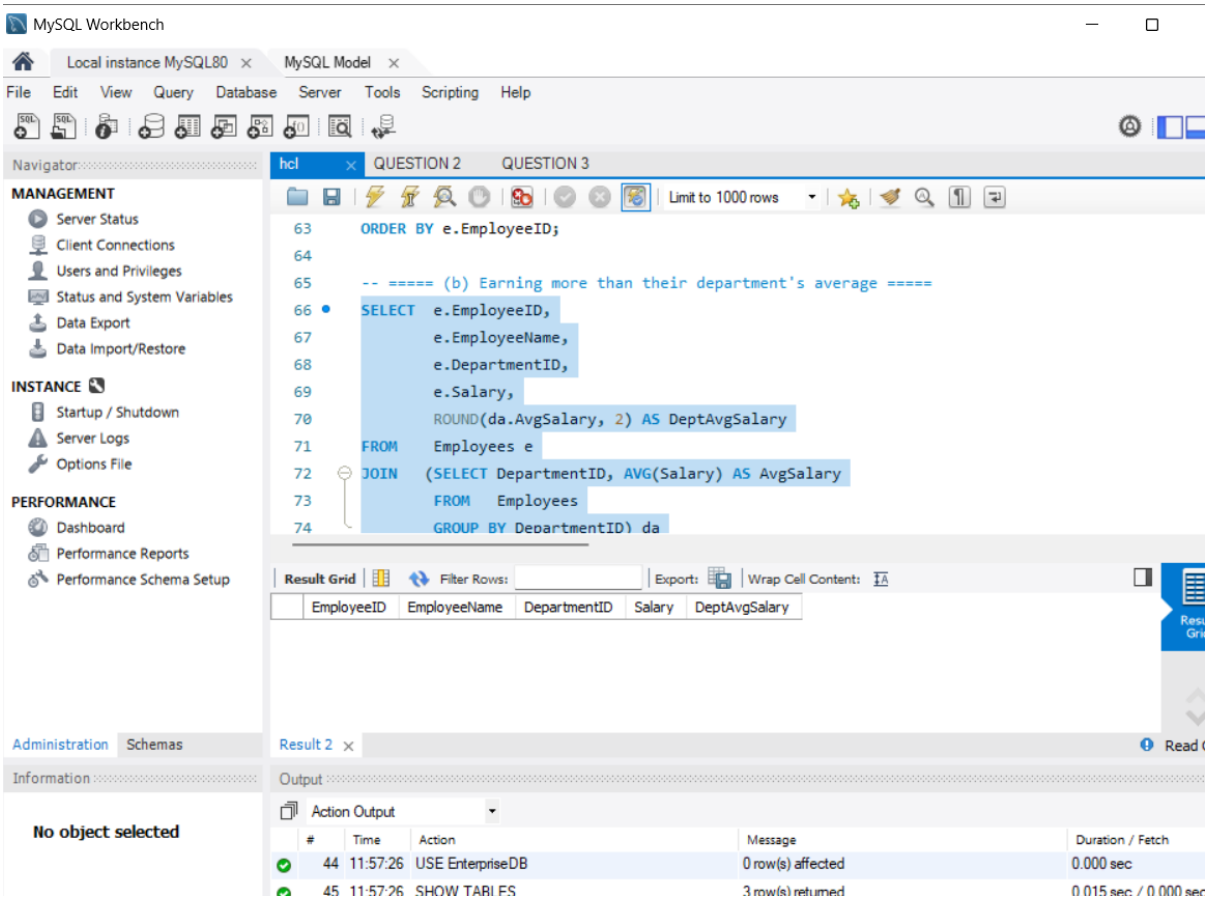

MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
MANAGEMENT
Server Status
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Users and Privileges
Status and System Variables
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INSTANCE
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Options File
PERFORMANCE
Dashboard
Performance Reports
Performance Schema Setup
Administration Schemas
Information
No object selected
Query 1
Limit to 1000 rows
13);
14
15 CREATE TABLE Employees (
16 EmployeeID INT NOT NULL,
17 EmployeeName VARCHAR(100) NOT NULL,
18 DepartmentID INT NOT NULL,
19 Salary DECIMAL(10,2) NOT NULL,
20 JoiningDate DATE NOT NULL,
21 ManagerID INT NULL,
22 CONSTRAINT pk_employees PRIMARY KEY (EmployeeID),
23 CONSTRAINT chk_salary CHECK (Salary > 0),
24 CONSTRAINT fk_emp_dept FOREIGN KEY (DepartmentID)
25 REFERENCES Departments (DepartmentID),
26 CONSTRAINT fk_emp_manager FOREIGN KEY (ManagerID)
27 REFERENCES Employees (EmployeeID)
28);
29
30 -- ===== DATA =====
31 INSERT INTO Departments (DepartmentID, DepartmentName, Location) VALUES
32 (101, 'IT', 'Bengaluru'),
Output
Action Output
Time Action Message Duration / Fetch
1 10:58:03 DROP DATABASE IF EXISTS EnterpriseDB 0 row(s) affected 0.093 sec
2 10:58:19 CREATE DATABASE EnterpriseDB 1 row(s) affected 0.016 sec
3 10:58:26 USE EnterpriseDB 0 row(s) affected 0.000 sec
4 10:58:41 CREATE TABLE Departments (DepartmentID IN... 0 row(s) affected 0.078 sec
5 10:58:50 CREATE TABLE Employees (EmployeeID INT ... Error Code: 1064. You have an error in your SQL synt... 0.016 sec



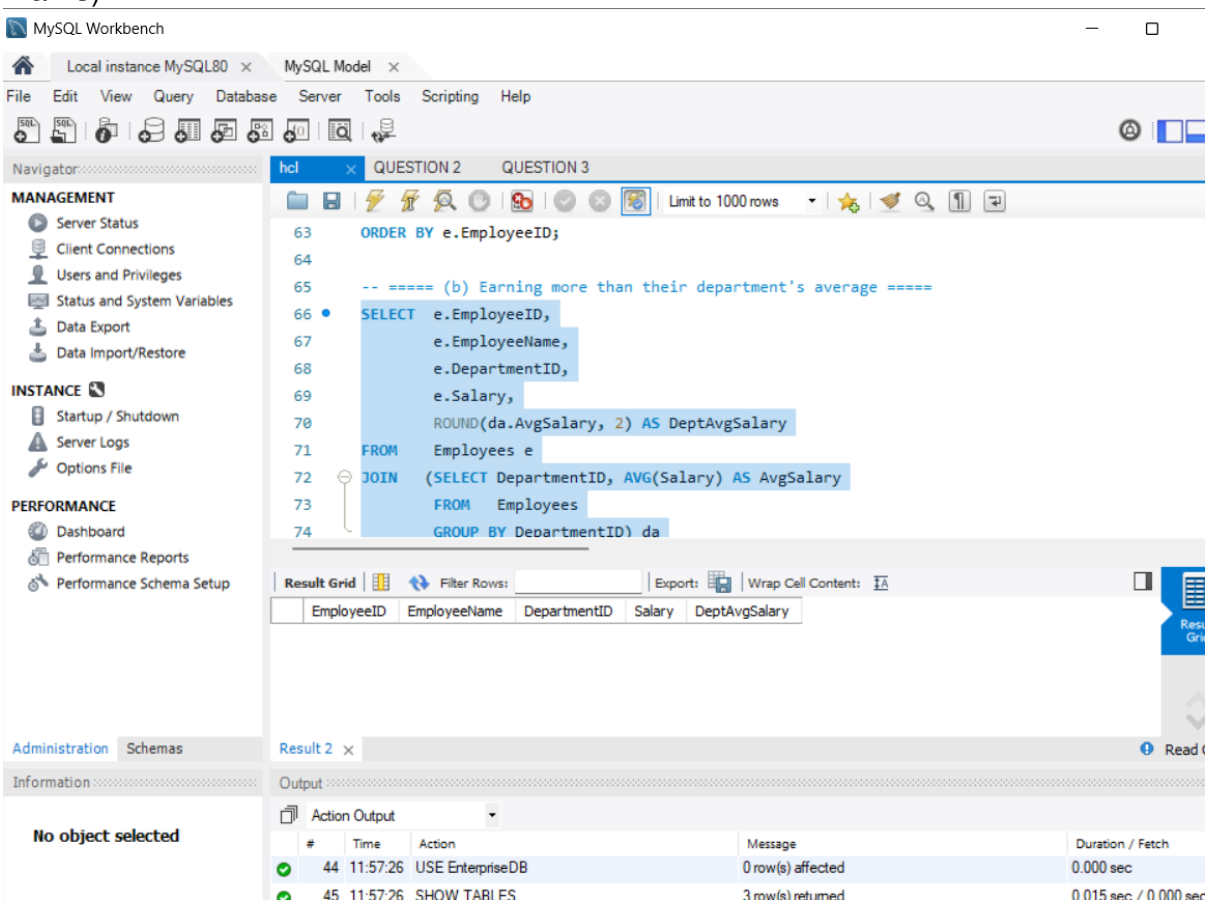
a)Display employee names, department names, and manager names using joins. (2 marks)



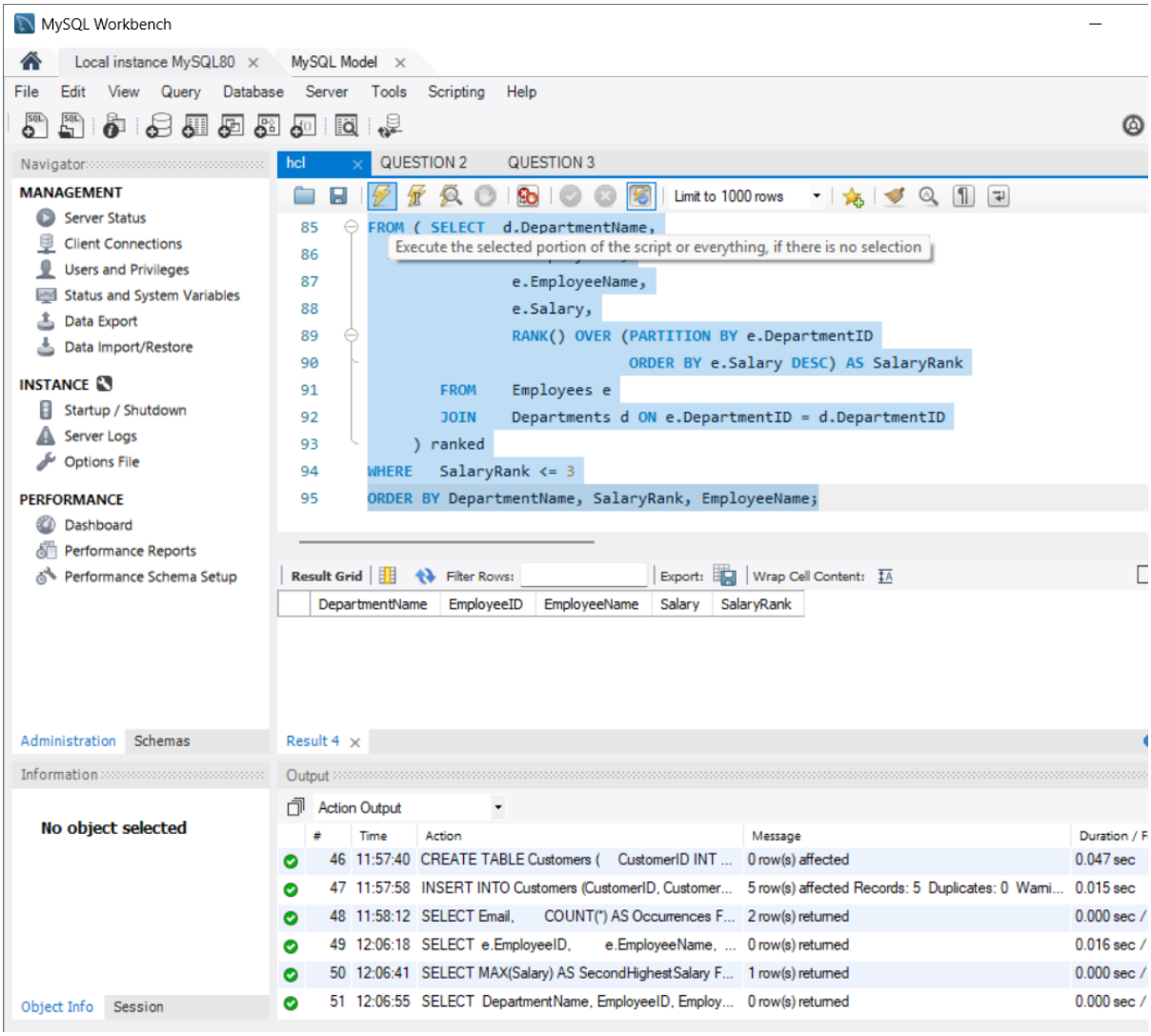
b. Find employees earning more than their department's average salary. (2 marks)



CFind the second-highest distinct salary without using LIMIT, TOP, or FETCH FIRST. (2 marks)

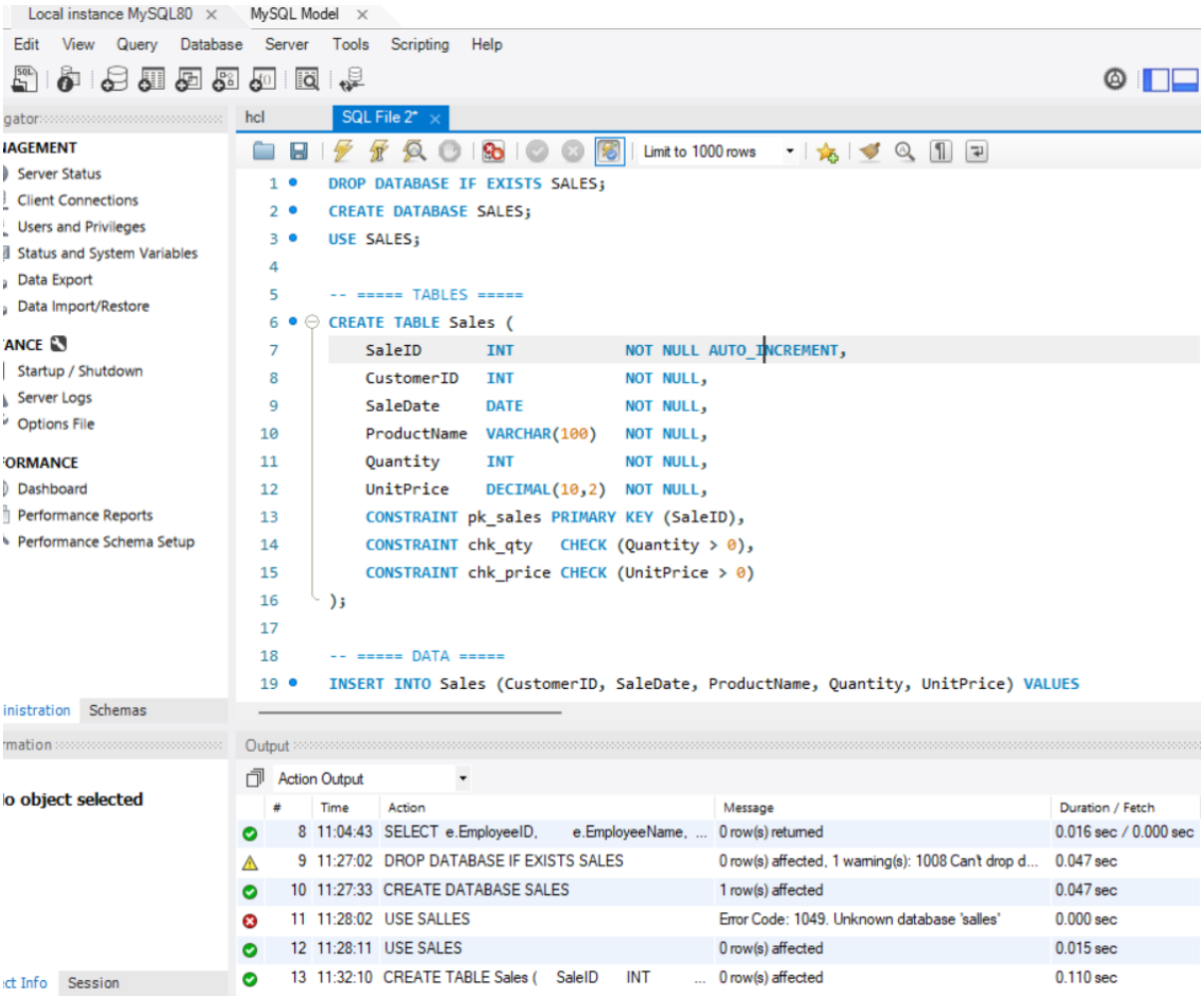


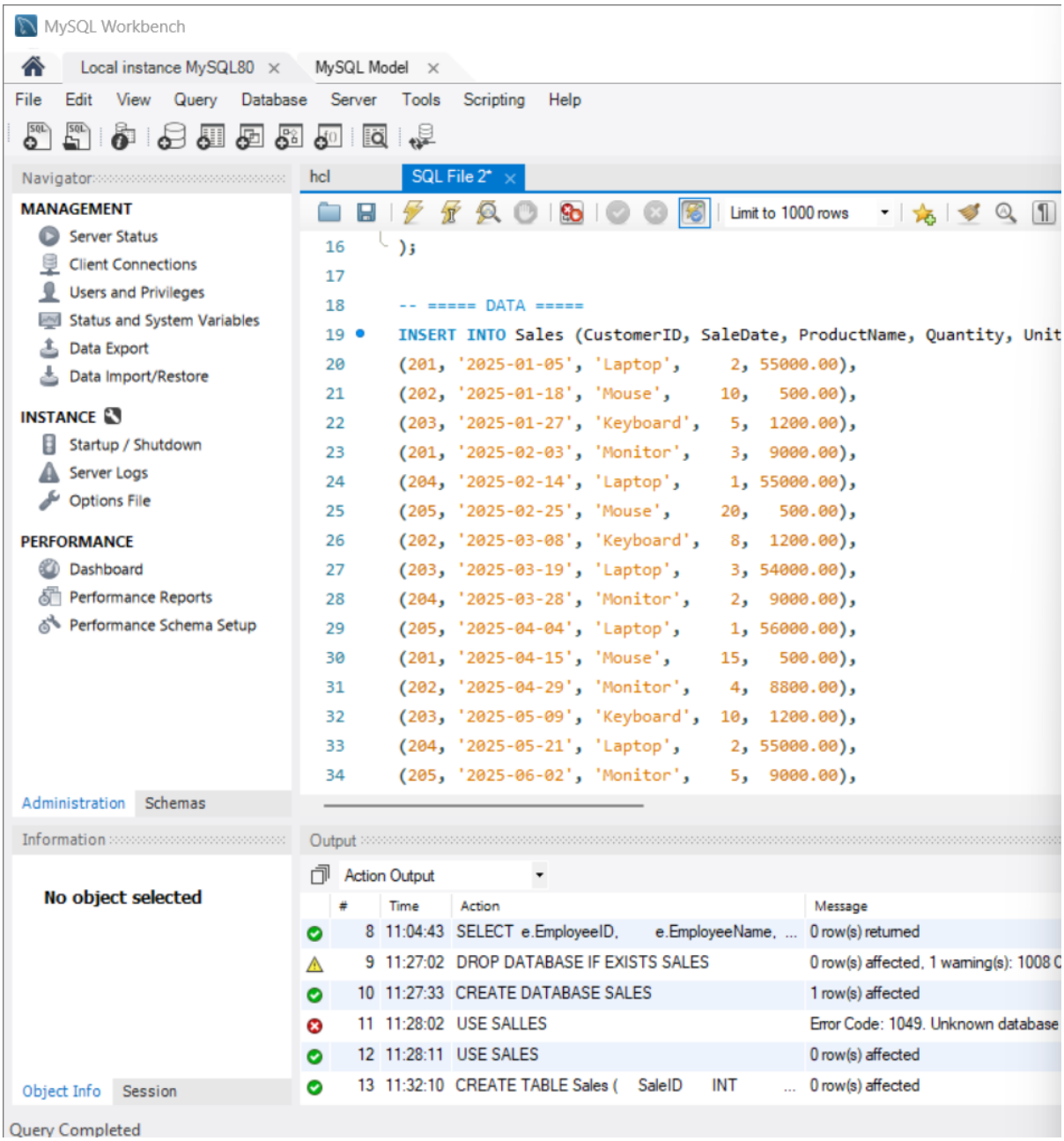
DRank employees by salary within each department and display the top three employees per department, retaining ties at the cutoff. (2 marks)



Question 2: Sales Analytics and Query Optimization (6 Marks)

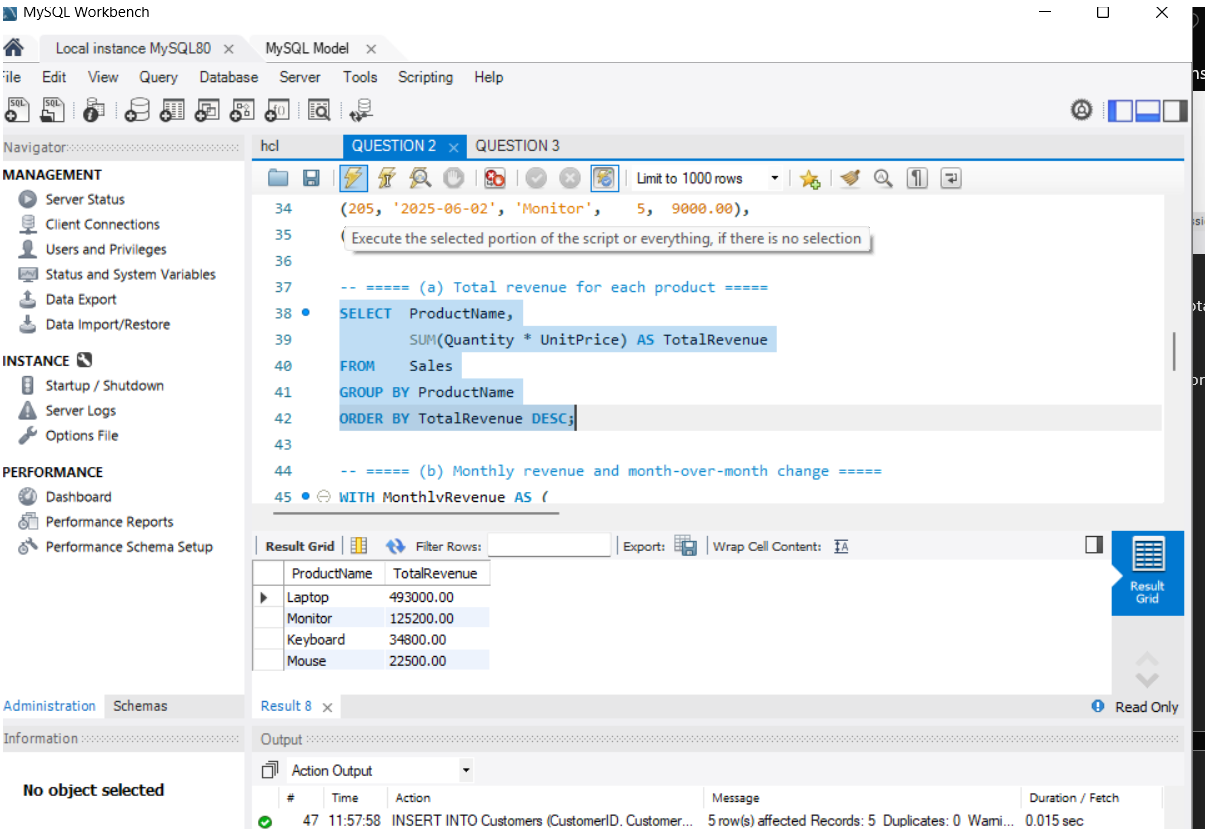
Question 2: Create a Sales table with the columns SaleID, CustomerID, SaleDate, ProductName, Quantity and UnitPrice. Insert at least 15 sample records covering multiple customers, products, and dates.


Local instance MySQL80
MySQL Model
Edit View Query Database Server Tools Scripting Help
hcl
SQL File 2*
Limit to 1000 rows
1 DROP DATABASE IF EXISTS SALES;
2 CREATE DATABASE SALES;
3 USE SALES;
4
5 -- ===== TABLES =====
6 CREATE TABLE Sales (
7 SaleID INT NOT NULL AUTO_INCREMENT,
8 CustomerID INT NOT NULL,
9 SaleDate DATE NOT NULL,
10 ProductName VARCHAR(100) NOT NULL,
11 Quantity INT NOT NULL,
12 UnitPrice DECIMAL(10,2) NOT NULL,
13 CONSTRAINT pk_sales PRIMARY KEY (SaleID),
14 CONSTRAINT chk_qty CHECK (Quantity > 0),
15 CONSTRAINT chk_price CHECK (UnitPrice > 0)
16);
17
18 -- ===== DATA =====
19 INSERT INTO Sales (CustomerID, SaleDate, ProductName, Quantity, UnitPrice) VALUES
Server Status
Client Connections
Users and Privileges
Status and System Variables
Data Export
Data Import/Restore
Startup / Shutdown
Server Logs
Options File
Dashboard
Performance Reports
Performance Schema Setup
Schemas
No object selected
Session
Output
Action Output
Time Action Message Duration / Fetch
8 11:04:43 SELECT e.EmployeeID, e.EmployeeName, ... 0 row(s) returned 0.016 sec / 0.000 sec
9 11:27:02 DROP DATABASE IF EXISTS SALES 0 row(s) affected, 1 warning(s): 1008 Can't drop d... 0.047 sec
10 11:27:33 CREATE DATABASE SALES 1 row(s) affected 0.047 sec
11 11:28:02 USE SALLES Error Code: 1049. Unknown database 'salles' 0.000 sec
12 11:28:11 USE SALES 0 row(s) affected 0.015 sec
13 11:32:10 CREATE TABLE Sales (SaleID INT ... 0 row(s) affected 0.110 sec

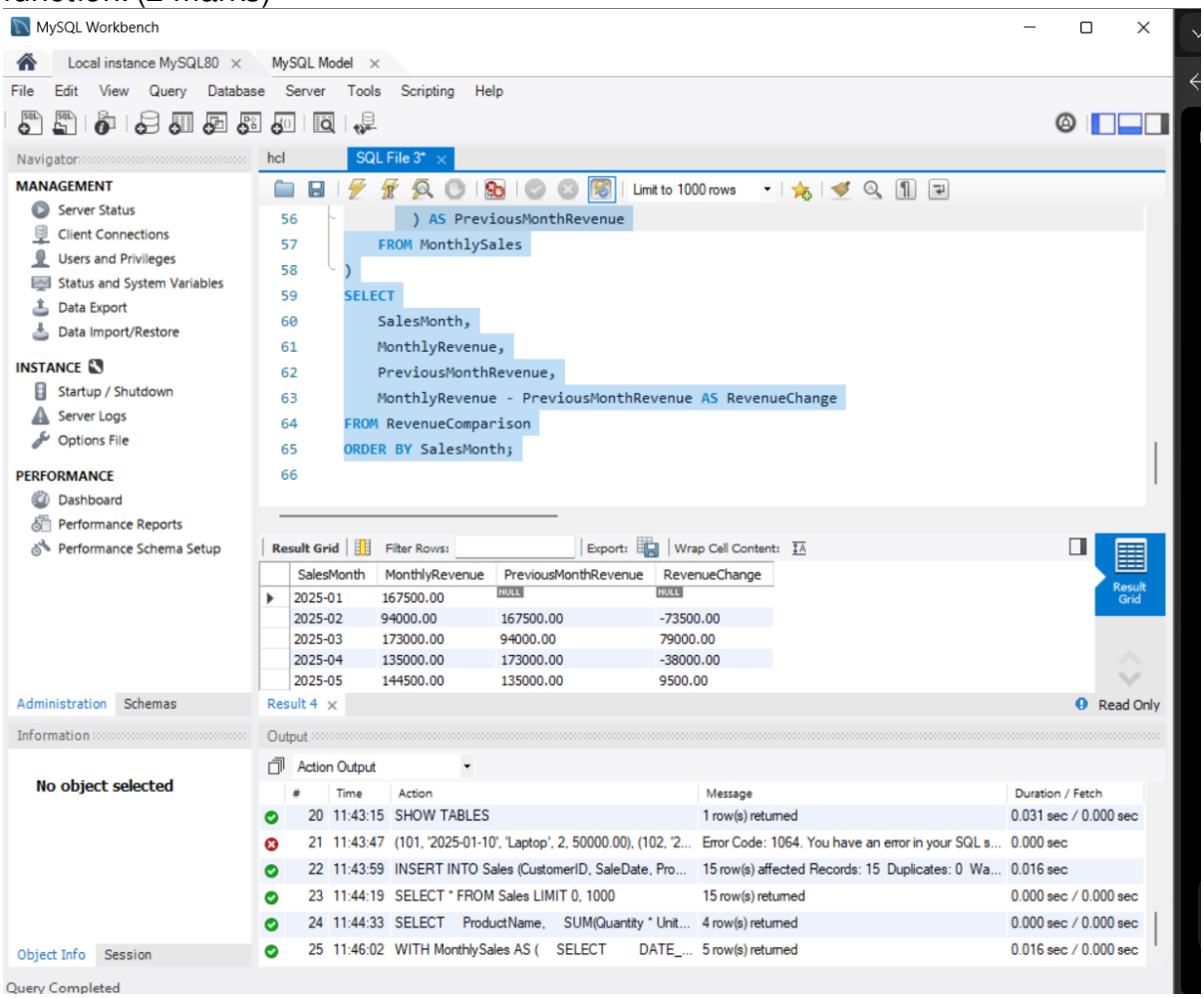


QUESTION 2

A)Calculate total revenue for each product. (2 marks)

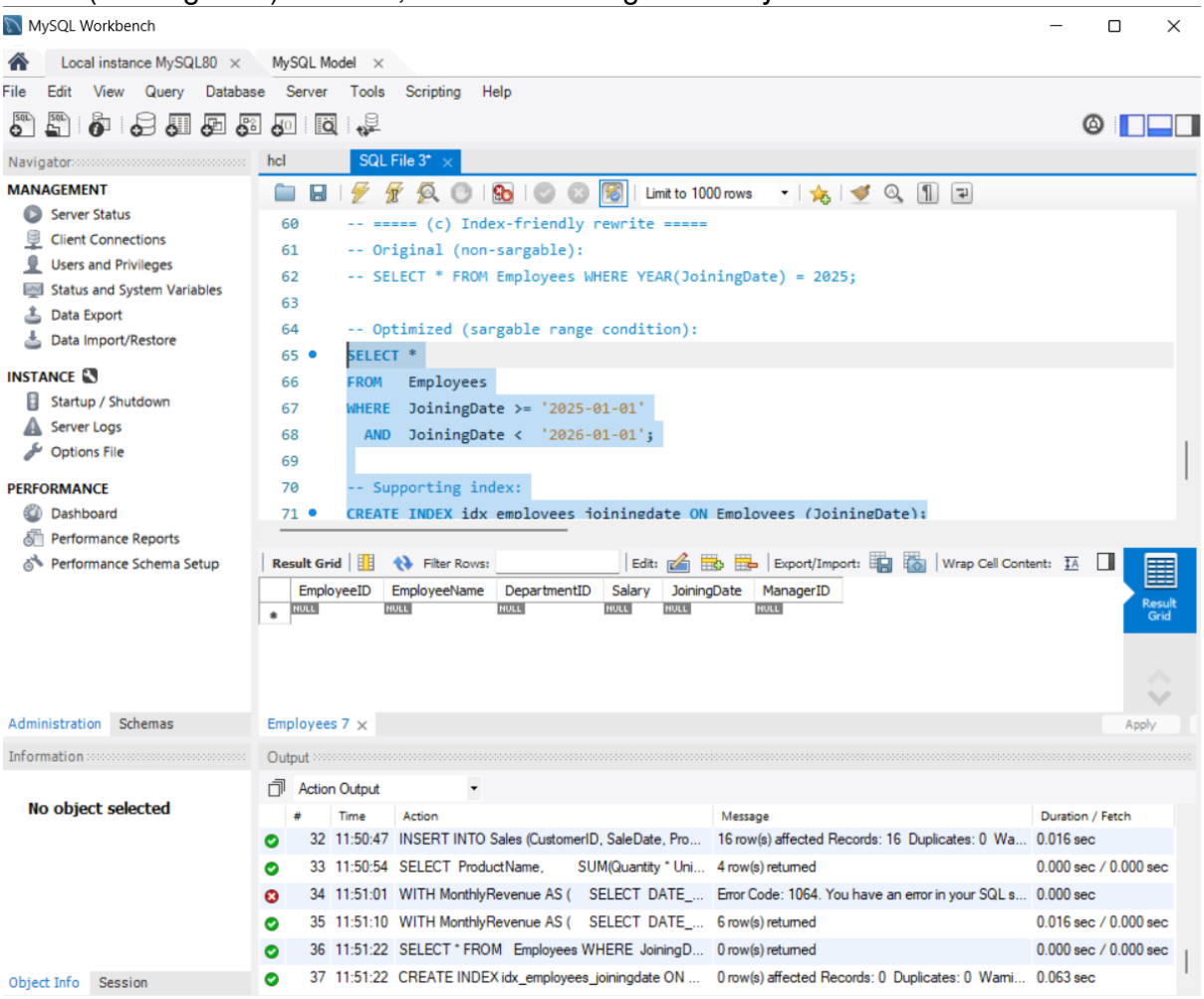


BCalculate monthly revenue and month-over-month revenue change using a window function. (2 marks)



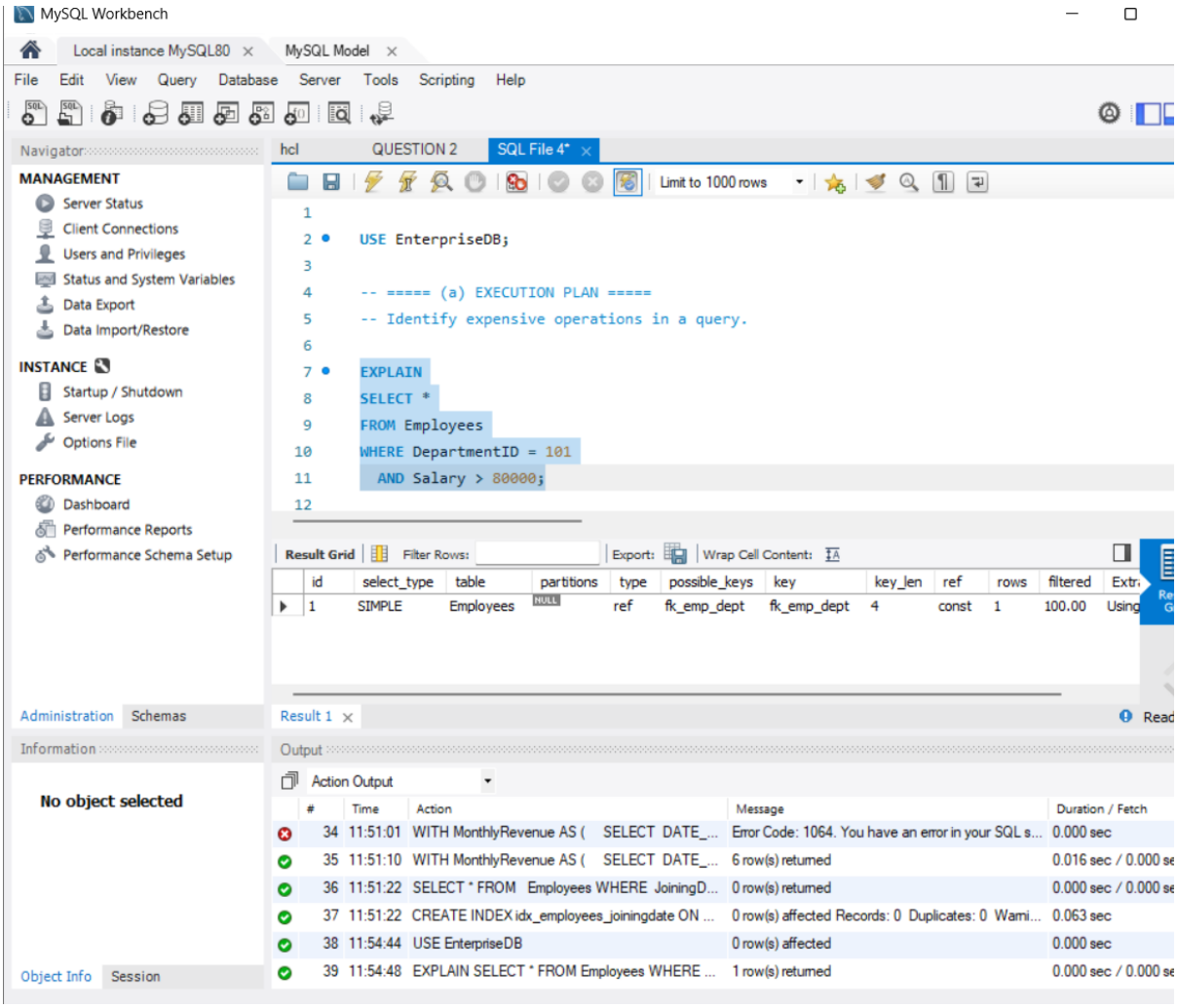
Cc. Rewrite the following query to improve the potential for index usage and explain why

your version may perform better. (2 marks) SELECT * FROM Employees WHERE YEAR(JoiningDate) = 2025; Assume JoiningDate may be indexed.

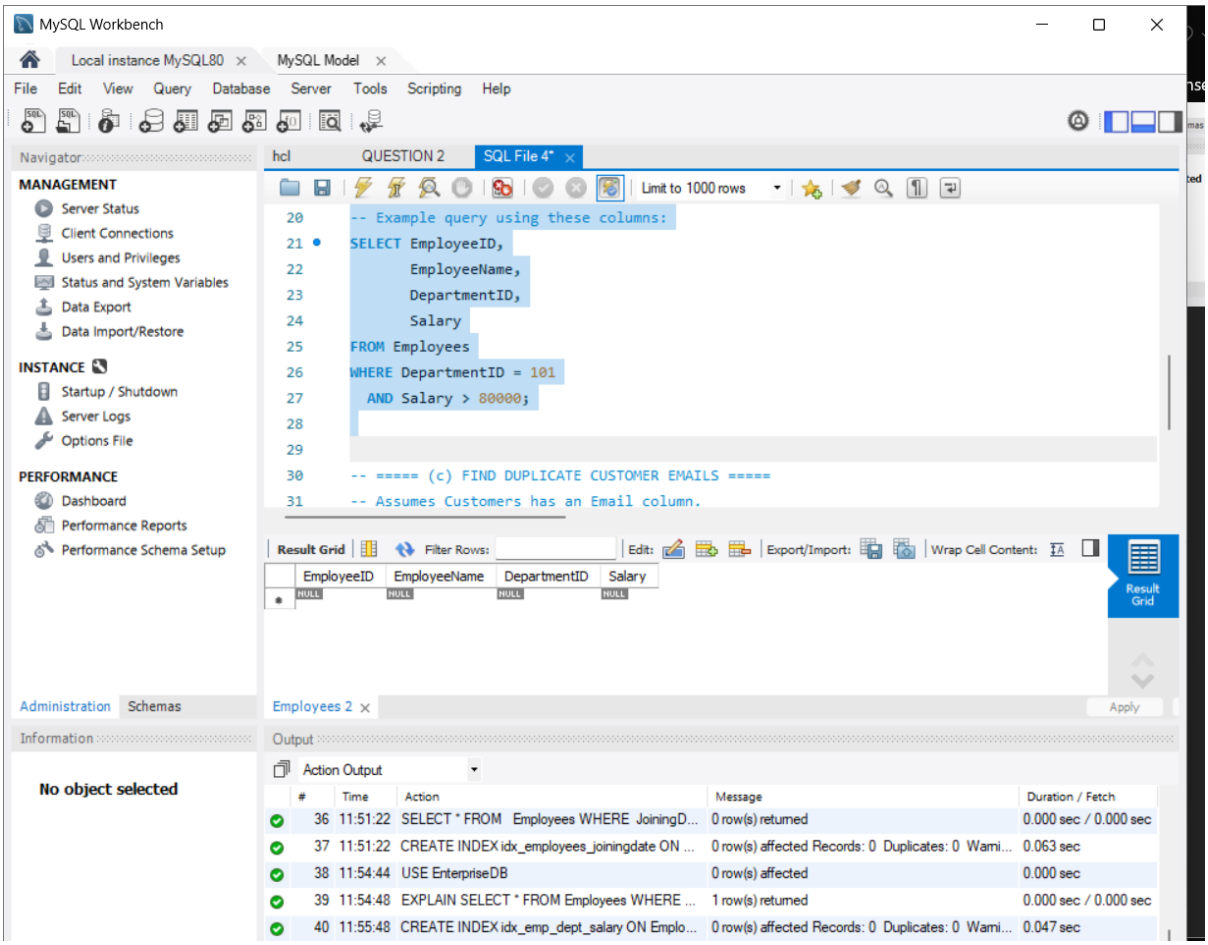


QUESTION 3

Question 3: SQL Troubleshooting (6 Marks) A company maintains millions of employee and transaction records. Its reporting queries are running slowly. Perform the following tasks: a. Explain how an execution plan helps identify expensive operations in a query. (2 marks)

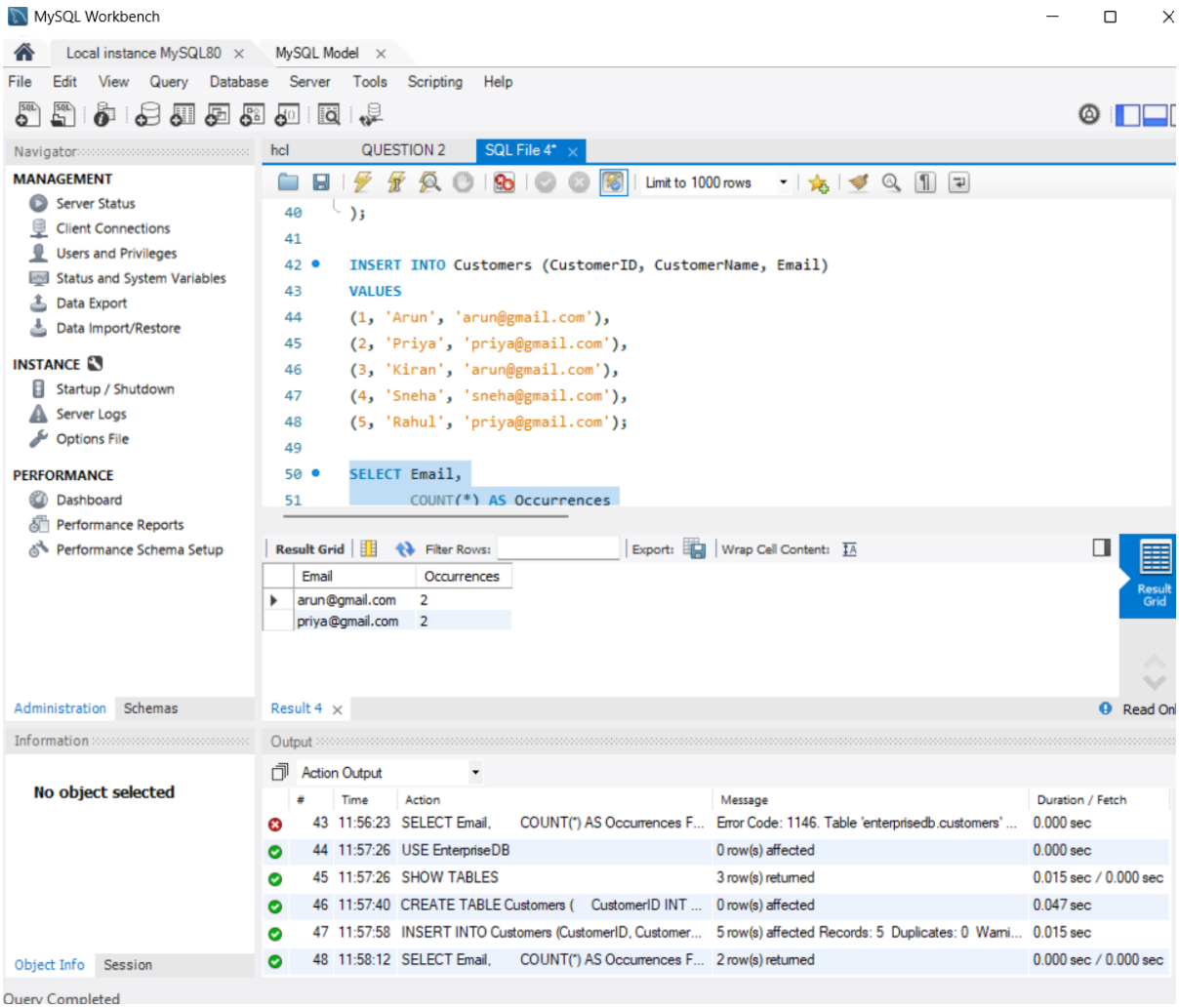


B) b. Recommend an appropriate index strategy for queries filtering employees by DepartmentID and Salary. Explain your reasoning. (2 marks)



C

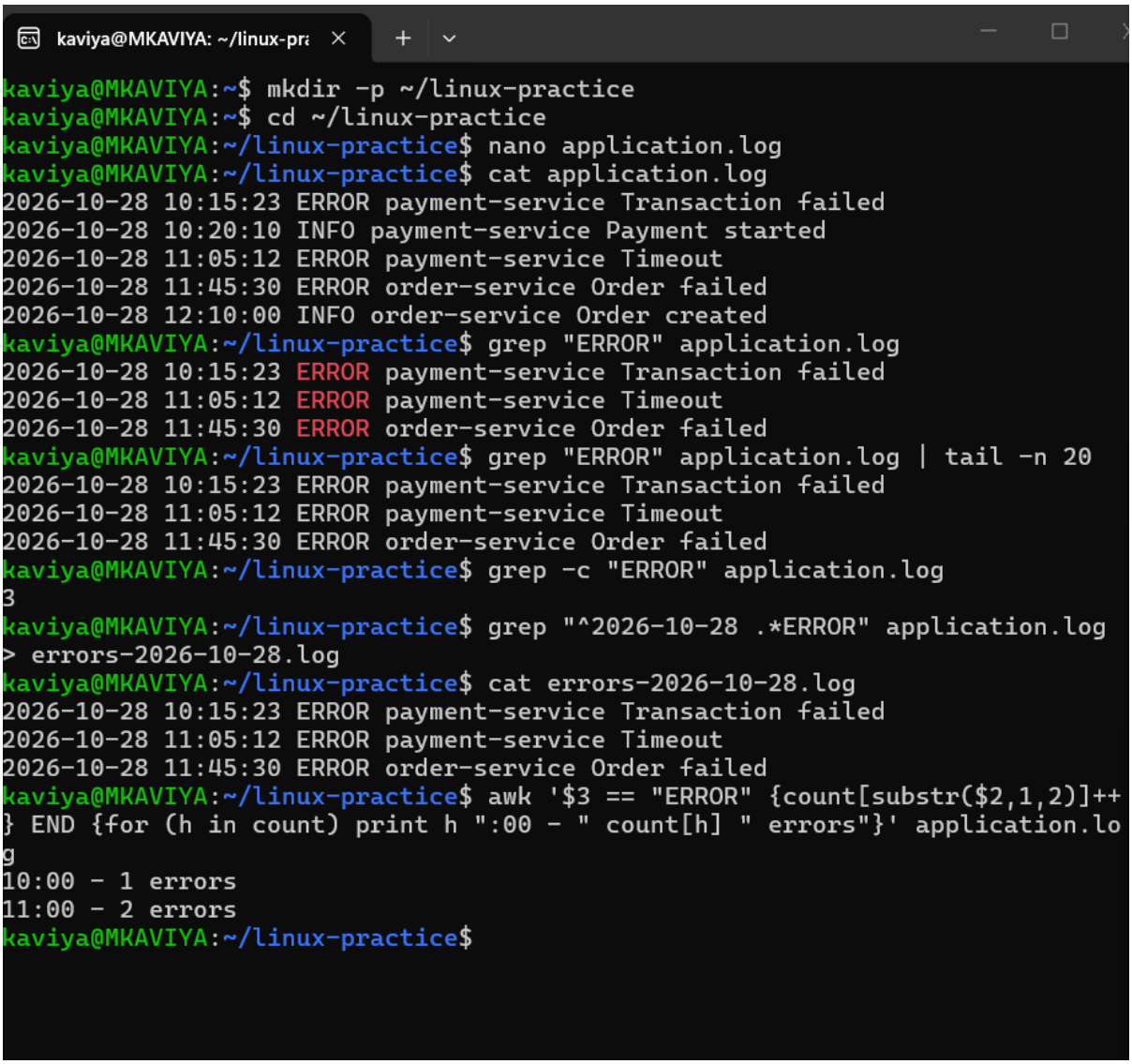
Write a query to identify duplicate customer email addresses and display only addresses occurring more than once. (2 marks)

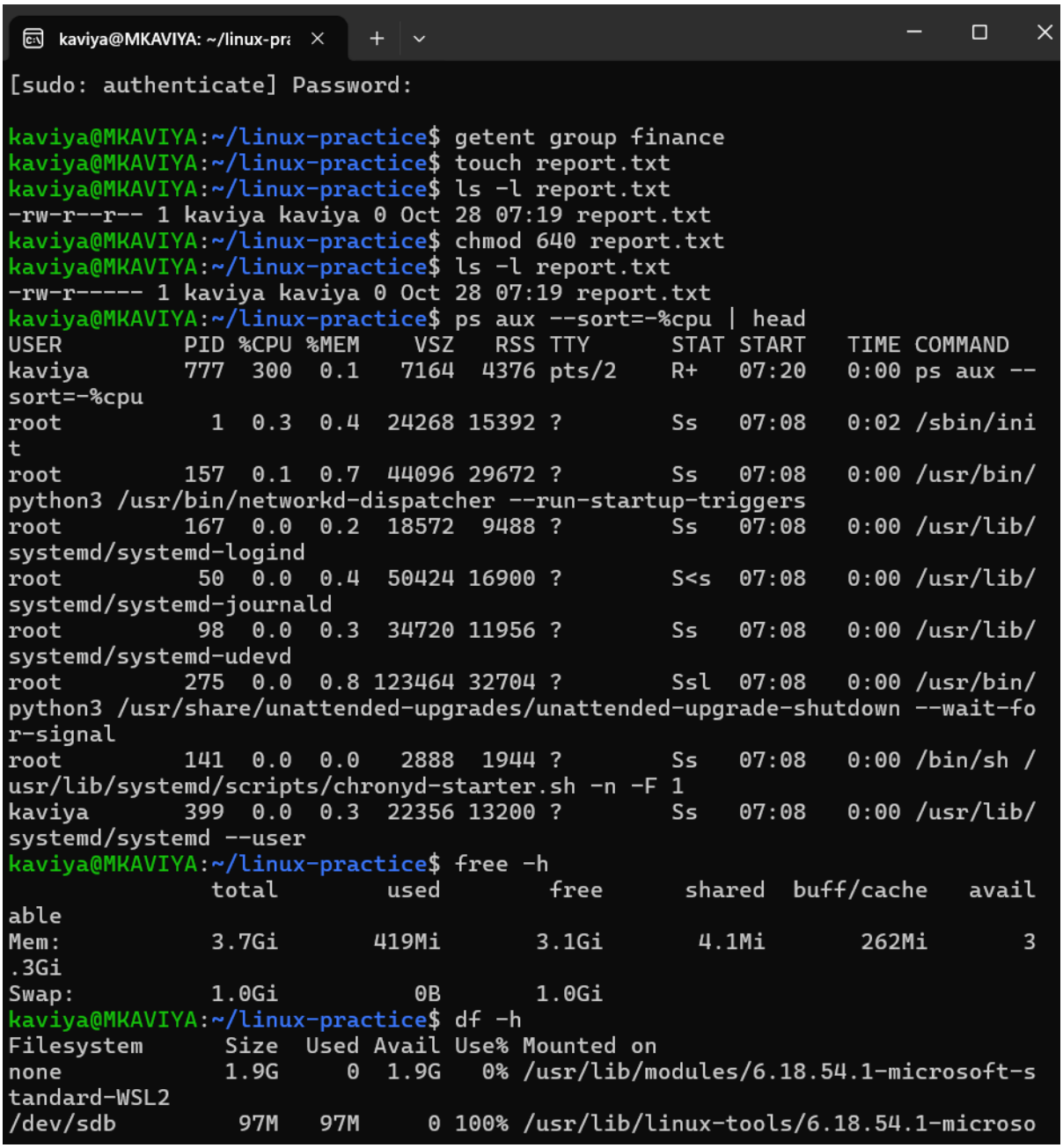


LINUX

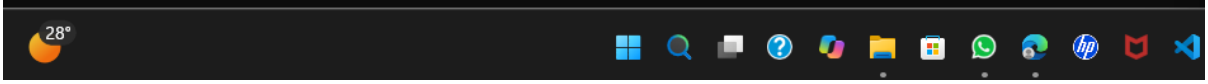
A

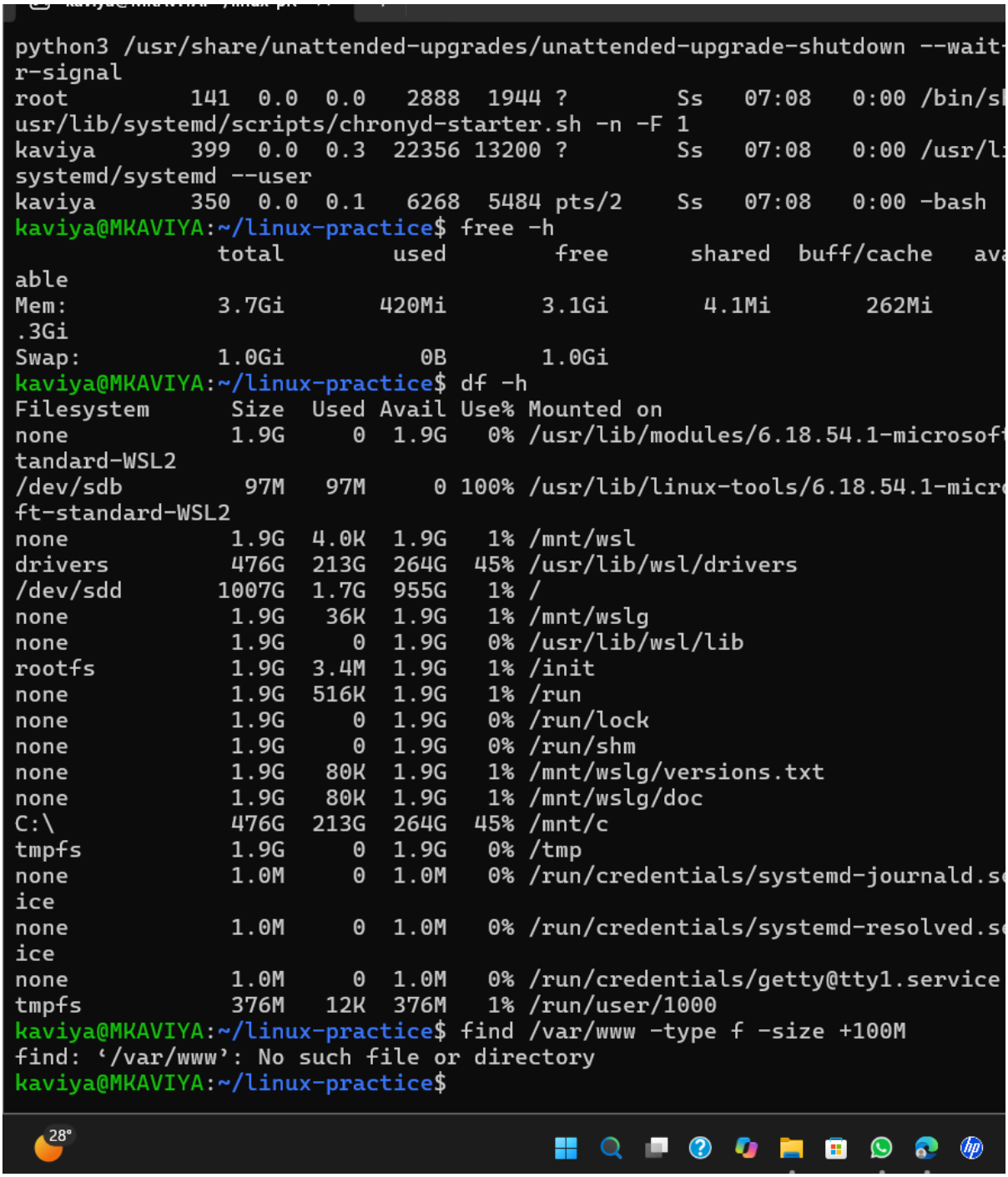
Question 4: Log Analysis (5 Marks)
Assume /var/log/application.log contains millions of log entries in the format: 2026-10-28 10:15:23 ERROR payment-service Transaction failed
Write Linux commands to:
a. Find all lines containing ERROR. (1 mark)
b. Display the 20 most recent error entries in the file. (1 mark)
c. Count the total number of error entries. (1 mark)
d. Extract error entries for a specified date into a separate file. (1 mark)
e. Explain how you would calculate the number of errors occurring in each hour. (1 mark)


kaviya@MKAVIYA: ~/linux-pra
kaviya@MKAVIYA:~$ mkdir -p ~/linux-practice
kaviya@MKAVIYA:~$ cd ~/linux-practice
kaviya@MKAVIYA:~/linux-practice$ nano application.log
kaviya@MKAVIYA:~/linux-practice$ cat application.log
2026-10-28 10:15:23 ERROR payment-service Transaction failed
2026-10-28 10:20:10 INFO payment-service Payment started
2026-10-28 11:05:12 ERROR payment-service Timeout
2026-10-28 11:45:30 ERROR order-service Order failed
2026-10-28 12:10:00 INFO order-service Order created
kaviya@MKAVIYA:~/linux-practice$ grep "ERROR" application.log
2026-10-28 10:15:23 ERROR payment-service Transaction failed
2026-10-28 11:05:12 ERROR payment-service Timeout
2026-10-28 11:45:30 ERROR order-service Order failed
kaviya@MKAVIYA:~/linux-practice$ grep "ERROR" application.log | tail -n 20
2026-10-28 10:15:23 ERROR payment-service Transaction failed
2026-10-28 11:05:12 ERROR payment-service Timeout
2026-10-28 11:45:30 ERROR order-service Order failed
kaviya@MKAVIYA:~/linux-practice$ grep -c "ERROR" application.log
3
kaviya@MKAVIYA:~/linux-practice$ grep "^2026-10-28 .*ERROR" application.log
> errors-2026-10-28.log
kaviya@MKAVIYA:~/linux-practice$ cat errors-2026-10-28.log
2026-10-28 10:15:23 ERROR payment-service Transaction failed
2026-10-28 11:05:12 ERROR payment-service Timeout
2026-10-28 11:45:30 ERROR order-service Order failed
kaviya@MKAVIYA:~/linux-practice$ awk '$3 == "ERROR" {count[substr($2,1,2)]++
} END {for (h in count) print h ":00 - " count[h] " errors"}' application.lo
g
10:00 - 1 errors
11:00 - 2 errors
kaviya@MKAVIYA:~/linux-practice$


kaviya@MKAVIYA: ~/linux-pra
[sudo: authenticate] Password:
kaviya@MKAVIYA:~/linux-practice$ getent group finance
kaviya@MKAVIYA:~/linux-practice$ touch report.txt
kaviya@MKAVIYA:~/linux-practice$ ls -l report.txt
-rw-r--r-- 1 kaviya kaviya 0 Oct 28 07:19 report.txt
kaviya@MKAVIYA:~/linux-practice$ chmod 640 report.txt
kaviya@MKAVIYA:~/linux-practice$ ls -l report.txt
-rw-r----- 1 kaviya kaviya 0 Oct 28 07:19 report.txt
kaviya@MKAVIYA:~/linux-practice$ ps aux --sort=-%cpu | head
USER PID %CPU %MEM VSZ RSS TTY STAT START TIME COMMAND
kaviya 777 300 0.1 7164 4376 pts/2 R+ 07:20 0:00 ps aux --sort=-%cpu
root 1 0.3 0.4 24268 15392 ? Ss 07:08 0:02 /sbin/init
root 157 0.1 0.7 44096 29672 ? Ss 07:08 0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
root 167 0.0 0.2 18572 9488 ? Ss 07:08 0:00 /usr/lib/systemd/systemd-logind
root 50 0.0 0.4 50424 16900 ? S<s 07:08 0:00 /usr/lib/systemd/systemd-journald
root 98 0.0 0.3 34720 11956 ? Ss 07:08 0:00 /usr/lib/systemd/systemd-udevd
root 275 0.0 0.8 123464 32704 ? Ssl 07:08 0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root 141 0.0 0.0 2888 1944 ? Ss 07:08 0:00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
kaviya 399 0.0 0.3 22356 13200 ? Ss 07:08 0:00 /usr/lib/systemd/systemd --user
kaviya@MKAVIYA:~/linux-practice$ free -h
total used free shared buff/cache available
Mem: 3.7Gi 419Mi 3.1Gi 4.1Mi 262Mi 3.3Gi
Swap: 1.0Gi 0B 1.0Gi
kaviya@MKAVIYA:~/linux-practice$ df -h
Filesystem Size Used Avail Use% Mounted on
none 1.9G 0 1.9G 0% /usr/lib/modules/6.18.54.1-microsoft-standard-WSL2
/dev/sdb 97M 97M 0 100% /usr/lib/linux-tools/6.18.54.1-microso

```
Filesystem      Size  Used Avail Use% Mounted on
none            1.9G     0  1.9G   0% /usr/lib/modules/6.18.54.1-microsoft-s
tandard-WSL2
/dev/sdb         97M   97M     0 100% /usr/lib/linux-tools/6.18.54.1-microso
ft-standard-WSL2
none            1.9G  4.0K  1.9G   1% /mnt/wsl
drivers         476G  213G  264G  45% /usr/lib/wsl/drivers
/dev/sdd       1007G  1.7G  955G   1% /
none            1.9G   36K  1.9G   1% /mnt/wslg
none            1.9G     0  1.9G   0% /usr/lib/wsl/lib
rootfs          1.9G  3.4M  1.9G   1% /init
none            1.9G  516K  1.9G   1% /run
none            1.9G     0  1.9G   0% /run/lock
none            1.9G     0  1.9G   0% /run/shm
none            1.9G   80K  1.9G   1% /mnt/wslg/versions.txt
none            1.9G   80K  1.9G   1% /mnt/wslg/doc
C:\             476G  213G  264G  45% /mnt/c
tmpfs           1.9G     0  1.9G   0% /tmp
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.serv
ice
none            1.0M     0  1.0M   0% /run/credentials/systemd-resolved.serv
ice
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           376M   12K  376M   1% /run/user/1000
kaviya@MKAVIYA:~/linux-practice$ find /var/www -type f -size +100M
find: ‘/var/www’: No such file or directory
kaviya@MKAVIYA:~/linux-practice$ find ~ -type f -size +100M
kaviya@MKAVIYA:~/linux-practice$ ls -l report.txt
-rw-r----- 1 kaviya kaviya 0 Oct 28 07:19 report.txt
kaviya@MKAVIYA:~/linux-practice$ ps aux --sort=-%cpu | head
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.3  0.4  24268 15392 ?        Ss   07:08   0:02 /sbin/ini
t
root         157  0.0  0.7  44096 29672 ?        Ss   07:08   0:00 /usr/bin/
python3 /usr/bin/networkd-dispatcher --run-startup-triggers
root         167  0.0  0.2  18572  9488 ?        Ss   07:08   0:00 /usr/lib/
systemd/systemd-logind
root          50  0.0  0.4  50424 16900 ?        S<s  07:08   0:00 /usr/lib/
systemd/systemd-journald
root          98  0.0  0.3  34720 11956 ?        Ss   07:08   0:00 /usr/lib/
systemd/systemd-udevd
```





SECTION C: PYSPARK DATA ENGINEERING || Maximum Marks: 20

Question 7: PySpark DataFrame Operations (6 Marks)
Create a PySpark DataFrame containing EmployeeID, EmployeeName, Department, Salary and JoiningDate, with at least 15 sample records. Write and execute a PySpark program to:
a. Display the schema and first five records. (1 mark)
b. Filter employees whose monthly salary is greater than ₹60,000. (1 mark)
c. Add an AnnualSalary column calculated from the monthly salary. (1 mark)
d. Group employees by department and calculate the average salary and employee count. (2 marks)
e. Display the five highest-paid employees. (1 mark)


Command Prompt
version 4.2.0
Using Scala version 2.13.18, OpenJDK 64-Bit Server VM, 17.0.20.1
Branch HEAD
Compiled by user hgao on 2026-07-11T15:29:54Z
Revision 32f7299601108917fb01920a54e084595b7b3bf8
Url https://github.com/apache/spark
Type --help for more information.
C:\Users\kaviy>notepad EmployeeDataEngineering.py
C:\Users\kaviy>notepad EmployeeDataEngineering.py
C:\Users\kaviy>python EmployeeDataEngineering.py
WARNING: Using incubator modules: jdk.incubator.vector
26/10/28 13:17:48 WARN Shell: Did not find winutils.exe: java.io.FileNotFou
dException: java.io.FileNotFoundException: HADOOP_HOME and hadoop.home.dir
re unset. -see https://cwiki.apache.org/confluence/display/HADOOP2/WindowsP
oblems
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.prope
ties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLo
Level(newLevel).
26/10/28 13:17:53 WARN NativeCodeLoader: Unable to load native-hadoop libra
y for your platform... using builtin-java classes where applicable
(a) SCHEMA
root
|-- EmployeeID: long (nullable = true)
|-- EmployeeName: string (nullable = true)
|-- Department: string (nullable = true)
|-- Salary: long (nullable = true)
|-- JoiningDate: date (nullable = true)
First five records:
[Stage 0:>
(0 + 1)

```
[Stage 1:>                                                          (0 + 4)

+----------+------------+----------+------+-----------+
|EmployeeID|EmployeeName|Department|Salary|JoiningDate|
+----------+------------+----------+------+-----------+
|101       |Arun        |IT        |75000 |2022-01-10 |
|102       |Priya       |HR        |45000 |2021-03-15 |
|103       |Rahul       |IT        |85000 |2020-07-20 |
|104       |Divya       |Finance   |65000 |2023-02-01 |
|105       |Kiran       |HR        |55000 |2022-06-12 |
+----------+------------+----------+------+-----------+
only showing top 5 rows

(b) SALARY GREATER THAN 60000
[Stage 2:>                                                          (0 + 1)

[Stage 3:>                                                          (0 + 4)
[Stage 3:=============================================>             (3 + 1)

[Stage 4:>                                                          (0 + 3)

+----------+------------+----------+------+-----------+
|EmployeeID|EmployeeName|Department|Salary|JoiningDate|
+----------+------------+----------+------+-----------+
|101       |Arun        |IT        |75000 |2022-01-10 |
|103       |Rahul       |IT        |85000 |2020-07-20 |
|104       |Divya       |Finance   |65000 |2023-02-01 |
|106       |Sneha       |IT        |95000 |2019-11-05 |
|107       |Vijay       |Finance   |70000 |2021-09-18 |
|110       |Meena       |Finance   |80000 |2018-08-14 |
|111       |Ajay        |IT        |72000 |2022-10-30 |
|112       |Lakshmi     |HR        |62000 |2021-05-22 |
|114       |Geetha      |IT        |90000 |2020-03-08 |
|115       |Suresh      |Finance   |68000 |2019-06-19 |
+----------+------------+----------+------+-----------+


(c) ANNUAL SALARY
[Stage 5:>                                                          (0 + 1)
```

```
(c) ANNUAL SALARY
[Stage 5:>                                                          (0 + 1)

[Stage 6:>                                                          (0 + 4)
[Stage 6:=============================================>             (3 + 1)

[Stage 7:>                                                          (0 + 3)

+----------+------------+------+------------+
|EmployeeID|EmployeeName|Salary|AnnualSalary|
+----------+------------+------+------------+
|101       |Arun        |75000 |900000      |
|102       |Priya       |45000 |540000      |
|103       |Rahul       |85000 |1020000     |
|104       |Divya       |65000 |780000      |
|105       |Kiran       |55000 |660000      |
|106       |Sneha       |95000 |1140000     |
|107       |Vijay       |70000 |840000      |
|108       |Anu         |60000 |720000      |
|109       |Ravi        |48000 |576000      |
|110       |Meena       |80000 |960000      |
|111       |Ajay        |72000 |864000      |
|112       |Lakshmi     |62000 |744000      |
|113       |Manoj       |58000 |696000      |
|114       |Geetha      |90000 |1080000     |
|115       |Suresh      |68000 |816000      |
+----------+------------+------+------------+


(d) DEPARTMENT-WISE SUMMARY
[Stage 8:>                                                          (0 + 8)
[Stage 8:=================================================>       (7 + 1)

+----------+-------------+-------------+
|Department|AverageSalary|EmployeeCount|
+----------+-------------+-------------+
|IT        |79500.0      |6            |
|HR        |52500.0      |4            |
|Finance   |68200.0      |5            |
```

```
|113       |Manoj       |58000 |696000      |
|114       |Geetha      |90000 |1080000     |
|115       |Suresh      |68000 |816000      |
+----------+------------+------+------------+


(d) DEPARTMENT-WISE SUMMARY
[Stage 8:>                                                          (0 + 8)
[Stage 8:=================================================>       (7 + 1)

+----------+-------------+-------------+
|Department|AverageSalary|EmployeeCount|
+----------+-------------+-------------+
|IT        |79500.0      |6            |
|HR        |52500.0      |4            |
|Finance   |68200.0      |5            |
+----------+-------------+-------------+


(e) TOP FIVE HIGHEST-PAID EMPLOYEES
[Stage 11:>                                                         (0 + 8)
[Stage 11:==============>                                          (2 + 6)

+----------+------------+----------+------+-----------+------------+
|EmployeeID|EmployeeName|Department|Salary|JoiningDate|AnnualSalary|
+----------+------------+----------+------+-----------+------------+
|106       |Sneha       |IT        |95000 |2019-11-05 |1140000     |
|114       |Geetha      |IT        |90000 |2020-03-08 |1080000     |
|103       |Rahul       |IT        |85000 |2020-07-20 |1020000     |
|110       |Meena       |Finance   |80000 |2018-08-14 |960000      |
|101       |Arun        |IT        |75000 |2022-01-10 |900000      |
+----------+------------+----------+------+-----------+------------+
only showing top 5 rows

C:\Users\kaviy>SUCCESS: The process with PID 27396 (child process of PID 184
00) has been terminated.
SUCCESS: The process with PID 18400 (child process of PID 15336) has been te
rminated.
SUCCESS: The process with PID 15336 (child process of PID 27532) has been te
rminated.
cc
```

```
C:\Users\kaviy>pyspark --version
WARNING: Using incubator modules: jdk.incubator.vector
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 4.2.0
      /_/

Using Scala version 2.13.18, OpenJDK 64-Bit Server VM, 17.0.20.1
Branch HEAD
Compiled by user hgao on 2026-07-11T15:29:54Z
Revision 32f7299601108917fb01920a54e084595b7b3bf8
Url https://github.com/apache/spark
Type --help for more information.

C:\Users\kaviy>cd %USERPROFILE%\Desktop
The system cannot find the path specified.

C:\Users\kaviy>notepad question8.py

C:\Users\kaviy>notepad transactions.csv

C:\Users\kaviy>python question8.py
WARNING: Using incubator modules: jdk.incubator.vector
26/10/28 13:29:57 WARN Shell: Did not find winutils.exe: java.io.FileNotFo
dException: java.io.FileNotFoundException: HADOOP_HOME and hadoop.home.dir
re unset. -see https://cwiki.apache.org/confluence/display/HADOOP2/Windows
oblems
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.prop
ties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setL
Level(newLevel).
26/10/28 13:29:59 WARN NativeCodeLoader: Unable to load native-hadoop libr
y for your platform... using builtin-java classes where applicable
Original Data:
+-------------+----------+---------------+-------+--------+---------+
|TransactionID|CustomerID|TransactionDate|Product|Quantity|UnitPrice|
+-------------+----------+---------------+-------+--------+---------+
```

```
y for your platform... using builtin-java classes where applicable
Original Data:
+-------------+----------+---------------+-------+--------+---------+
|TransactionID|CustomerID|TransactionDate|Product|Quantity|UnitPrice|
+-------------+----------+---------------+-------+--------+---------+
|T101         |C101      |2026-10-20     |Laptop |2       |50000.0  |
|T102         |C102      |2026-10-21     |Mobile |3       |20000.0  |
|T103         |C103      |2026-10-22     |Laptop |1       |50000.0  |
|T101         |C101      |2026-10-20     |Laptop |2       |50000.0  |
|T104         |C104      |invalid-date   |Tablet |2       |15000.0  |
+-------------+----------+---------------+-------+--------+---------+

Cleaned Data:
[Stage 1:>                                                          (0 + 1)
[Stage 3:>                                                          (0 + 1)
26/10/28 13:30:12 ERROR Executor: Exception in task 0.0 in stage 3.0 (TID 2)
org.apache.spark.SparkDateTimeException: [CANNOT_PARSE_TIMESTAMP] Text 'inva
lid-date' could not be parsed at index 0. Use `try_to_date` to tolerate inva
lid input string and return NULL instead. SQLSTATE: 22007
        at org.apache.spark.sql.errors.QueryExecutionErrors$.ansiDateTimePar
seError(QueryExecutionErrors.scala:290)
        at org.apache.spark.sql.errors.QueryExecutionErrors.ansiDateTimePars
eError(QueryExecutionErrors.scala)
        at org.apache.spark.sql.catalyst.expressions.GeneratedClass$Generate
dIteratorForCodegenStage3.processNext(Unknown Source)
        at org.apache.spark.sql.execution.BufferedRowIterator.hasNext(Buffer
edRowIterator.java:44)
        at org.apache.spark.sql.execution.WholeStageCodegenEvaluatorFactory$
WholeStageCodegenPartitionEvaluator$$anon$1.hasNext(WholeStageCodegenEvaluat
orFactory.scala:50)
        at org.apache.spark.sql.execution.SparkPlan.$anonfun$getByteArrayRdd
$2(SparkPlan.scala:418)
        at org.apache.spark.rdd.RDD.$anonfun$mapPartitionsInternal$2(RDD.sca
la:910)
        at org.apache.spark.rdd.RDD.$anonfun$mapPartitionsInternal$2$adapted
(RDD.scala:910)
        at org.apache.spark.rdd.MapPartitionsRDD.compute(MapPartitionsRDD.sc
ala:57)
        at org.apache.spark.rdd.RDD.computeOrReadCheckpoint(RDD.scala:374)
        at org.apache.spark.rdd.RDD.iterator(RDD.scala:338)
```

```
C:\Users\kaviy>notepad question8.py

C:\Users\kaviy>python question8.py
WARNING: Using incubator modules: jdk.incubator.vector
26/10/28 13:32:07 WARN Shell: Did not find winutils.exe: java.io.FileNotF
dException: java.io.FileNotFoundException: HADOOP_HOME and hadoop.home.di
re unset. -see https://cwiki.apache.org/confluence/display/HADOOP2/Window
oblems
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.pro
ties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use set
Level(newLevel).
26/10/28 13:32:08 WARN NativeCodeLoader: Unable to load native-hadoop lib
y for your platform... using builtin-java classes where applicable
Original Data:
+-------------+----------+---------------+-------+--------+---------+
|TransactionID|CustomerID|TransactionDate|Product|Quantity|UnitPrice|
+-------------+----------+---------------+-------+--------+---------+
|T101         |C101      |2026-10-20     |Laptop |2       |50000.0  |
|T102         |C102      |2026-10-21     |Mobile |3       |20000.0  |
|T103         |C103      |2026-10-22     |Laptop |1       |50000.0  |
|T101         |C101      |2026-10-20     |Laptop |2       |50000.0  |
|T104         |C104      |invalid-date   |Tablet |2       |15000.0  |
+-------------+----------+---------------+-------+--------+---------+

Original Data:
+-------------+----------+---------------+-------+--------+---------+
|TransactionID|CustomerID|TransactionDate|Product|Quantity|UnitPrice|
+-------------+----------+---------------+-------+--------+---------+
|T101         |C101      |2026-10-20     |Laptop |2       |50000.0  |
|T102         |C102      |2026-10-21     |Mobile |3       |20000.0  |
|T103         |C103      |2026-10-22     |Laptop |1       |50000.0  |
|T101         |C101      |2026-10-20     |Laptop |2       |50000.0  |
|T104         |C104      |invalid-date   |Tablet |2       |15000.0  |
+-------------+----------+---------------+-------+--------+---------+

Cleaned Data:
```

```
|TransactionID|CustomerID|TransactionDate|Product|Quantity|UnitPrice|
+-------------+----------+---------------+-------+--------+---------+
|T101         |C101      |2026-10-20     |Laptop |2       |50000.0  |
|T102         |C102      |2026-10-21     |Mobile |3       |20000.0  |
|T103         |C103      |2026-10-22     |Laptop |1       |50000.0  |
|T101         |C101      |2026-10-20     |Laptop |2       |50000.0  |
|T104         |C104      |invalid-date   |Tablet |2       |15000.0  |
+-------------+----------+---------------+-------+--------+---------+

Cleaned Data:
+-------------+----------+---------------+-------+--------+---------+-------
-+
|TransactionID|CustomerID|TransactionDate|Product|Quantity|UnitPrice|Revenue
 |
+-------------+----------+---------------+-------+--------+---------+-------
-+
|T101         |C101      |2026-10-20     |Laptop |2       |50000.0  |100000.
0|
|T102         |C102      |2026-10-21     |Mobile |3       |20000.0  |60000.0
 |
|T103         |C103      |2026-10-22     |Laptop |1       |50000.0  |50000.0
 |
+-------------+----------+---------------+-------+--------+---------+-------
-+

Total Revenue by Product:
+-------+------------+
|Product|TotalRevenue|
+-------+------------+
|Laptop |150000.0    |
|Mobile |60000.0     |
+-------+------------+


C:\Users\kaviy>SUCCESS: The process with PID 1960 (child process of PID 3255
6) has been terminated.
SUCCESS: The process with PID 32556 (child process of PID 30768) has been te
rminated.
SUCCESS: The process with PID 30768 (child process of PID 11244) has been te
rminated.
```

Question 9: PySpark Joins and Performance (7 Marks)

Customers (CustomerID, CustomerName, City) and Orders (OrderID, CustomerID, OrderAmount, OrderDate) datasets. Write a PySpark program to:
a. Load both datasets into DataFrames. (1 mark)
b. Join the datasets to display orders with customer details. (2 marks)
c. Identify customers who have never placed an order. (1 mark)
d. Calculate the total order value by city. (1 mark)

```
C:\Users\kaviy>notepad question9.py

C:\Users\kaviy>notepad customers.csv

C:\Users\kaviy>notepad orders.csv

C:\Users\kaviy>python question9.py
WARNING: Using incubator modules: jdk.incubator.vector
26/10/28 13:37:15 WARN Shell: Did not find winutils.exe: java.io.FileNotFou
dException: java.io.FileNotFoundException: HADOOP_HOME and hadoop.home.dir
re unset. -see https://cwiki.apache.org/confluence/display/HADOOP2/WindowsP
oblems
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.prope
ties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLo
Level(newLevel).
26/10/28 13:37:16 WARN NativeCodeLoader: Unable to load native-hadoop libra
y for your platform... using builtin-java classes where applicable
Customers Data:
+----------+------------+---------+
|CustomerID|CustomerName|     City|
+----------+------------+---------+
|      C101|        Arun|Bangalore|
|      C102|       Priya|  Chennai|
|      C103|        Ravi|Hyderabad|
|      C104|       Meena|Bangalore|
+----------+------------+---------+

Orders Data:
+-------+----------+-----------+----------+
|OrderID|CustomerID|OrderAmount| OrderDate|
+-------+----------+-----------+----------+
|   O101|      C101|       2000|2026-10-20|
|   O102|      C102|       1500|2026-10-21|
|   O103|      C101|       3000|2026-10-22|
+-------+----------+-----------+----------+

Orders with Customer Details:
```

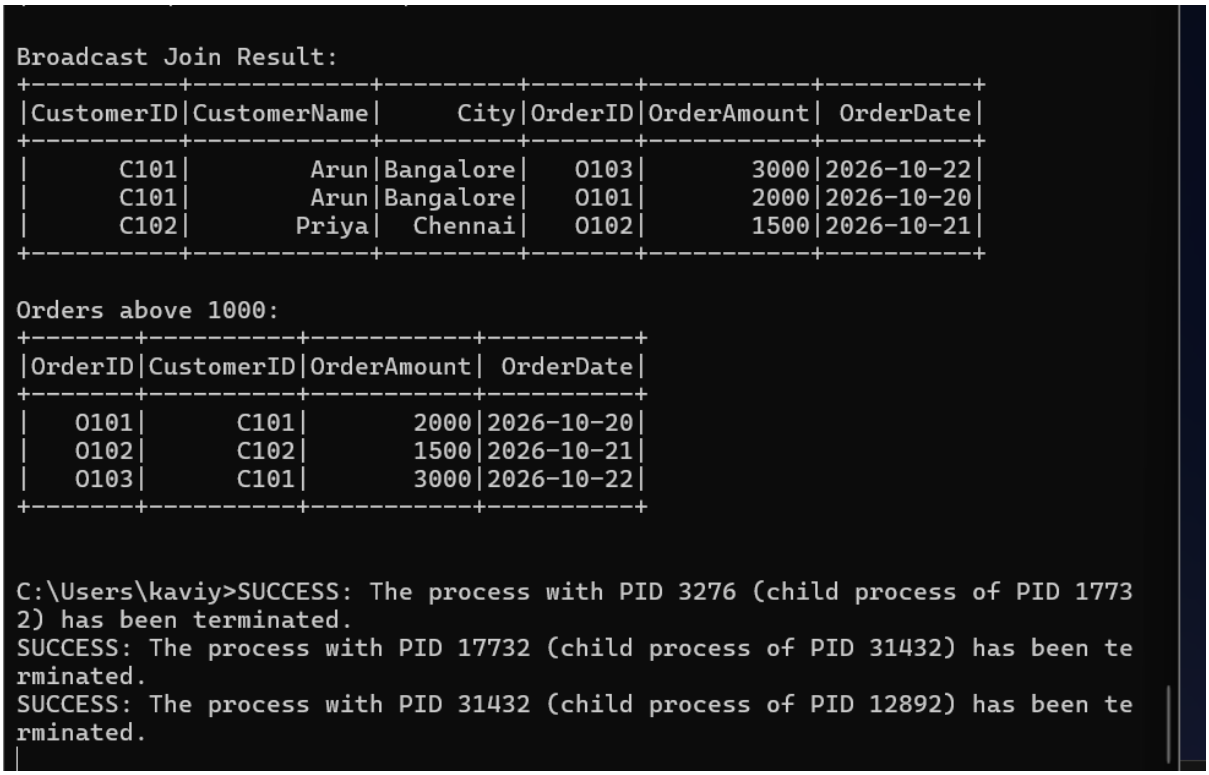
```
Orders with Customer Details:
+----------+------------+---------+-------+-----------+----------+
|CustomerID|CustomerName|     City|OrderID|OrderAmount| OrderDate|
+----------+------------+---------+-------+-----------+----------+
|      C101|        Arun|Bangalore|   O101|       2000|2026-10-20|
|      C102|       Priya|  Chennai|   O102|       1500|2026-10-21|
|      C101|        Arun|Bangalore|   O103|       3000|2026-10-22|
+----------+------------+---------+-------+-----------+----------+

Customers Who Never Placed Orders:
+----------+------------+---------+
|CustomerID|CustomerName|     City|
+----------+------------+---------+
|      C103|        Ravi|Hyderabad|
|      C104|       Meena|Bangalore|
+----------+------------+---------+

Total Order Value by City:
+---------+---------------+
|     City|TotalOrderValue|
+---------+---------------+
|Bangalore|           5000|
|  Chennai|           1500|
+---------+---------------+

Broadcast Join Result:
+----------+------------+---------+-------+-----------+----------+
|CustomerID|CustomerName|     City|OrderID|OrderAmount| OrderDate|
+----------+------------+---------+-------+-----------+----------+
|      C101|        Arun|Bangalore|   O103|       3000|2026-10-22|
|      C101|        Arun|Bangalore|   O101|       2000|2026-10-20|
|      C102|       Priya|  Chennai|   O102|       1500|2026-10-21|
+----------+------------+---------+-------+-----------+----------+

Orders above 1000:
+-------+----------+-----------+----------+
|OrderID|CustomerID|OrderAmount| OrderDate|
+-------+----------+-----------+----------+
|   O101|      C101|       2000|2026-10-20|
|   O102|      C102|       1500|2026-10-21|
```



(e) Question: Explain when a broadcast join is useful and what conditions make it appropriate. (1 Mark)

Answer: A broadcast join is useful when one DataFrame is small enough to fit in the memory of each worker. Spark distributes the small DataFrame to all workers, reducing data shuffling and improving join performance. It is appropriate when the smaller DataFrame fits safely in the available memory.

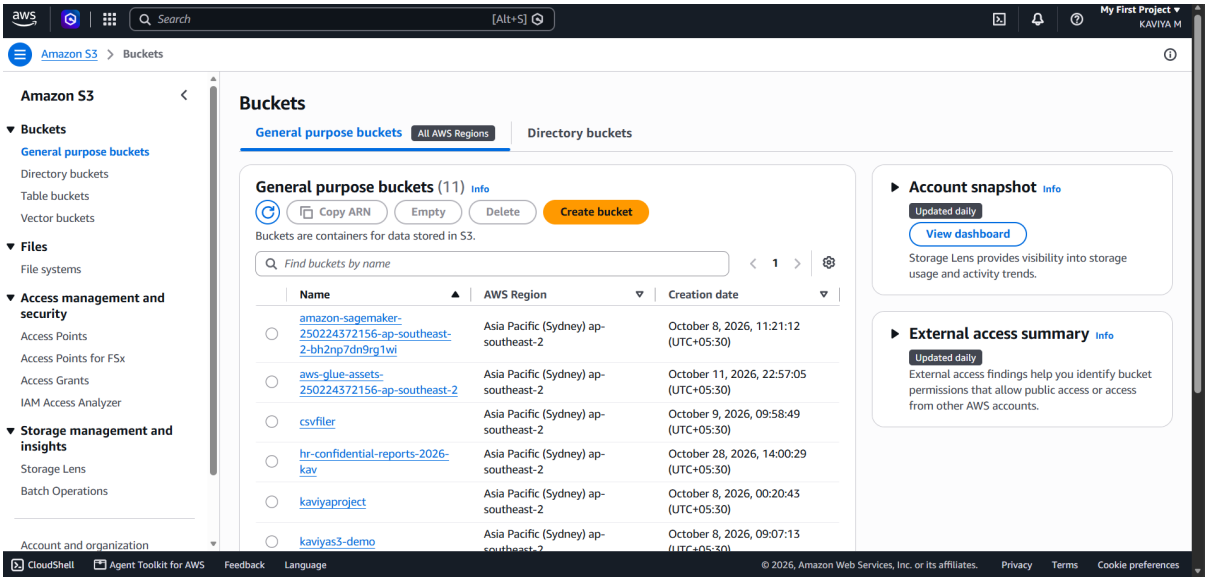
(f) Question: Explain the difference between transformations and actions, giving one example of each. (1 Mark)

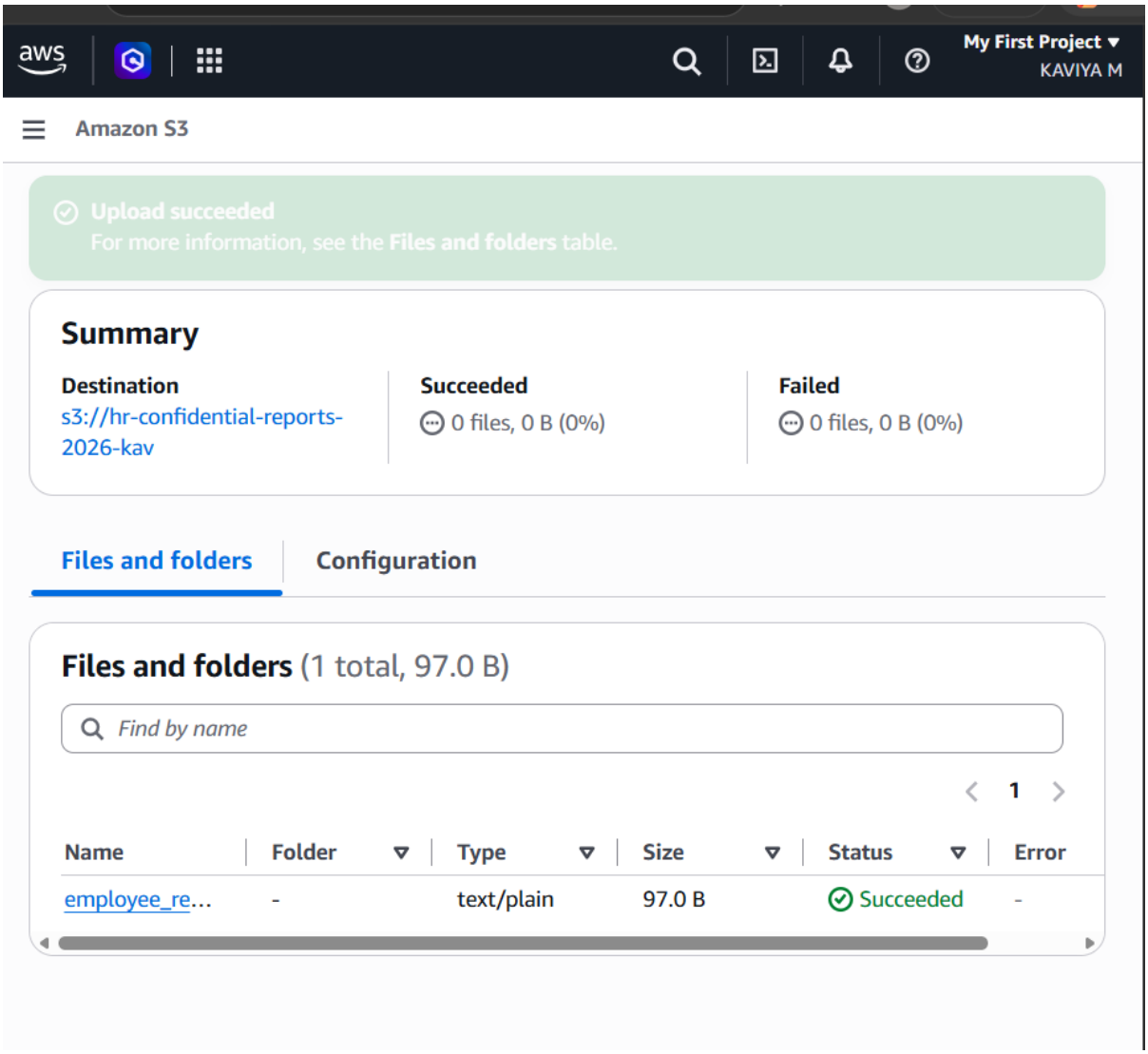
Answer: Transformations create a new DataFrame from an existing DataFrame and are evaluated lazily. For example, filter() selects rows that satisfy a condition. Actions trigger the execution of Spark computations. For example, show() displays the DataFrame results.

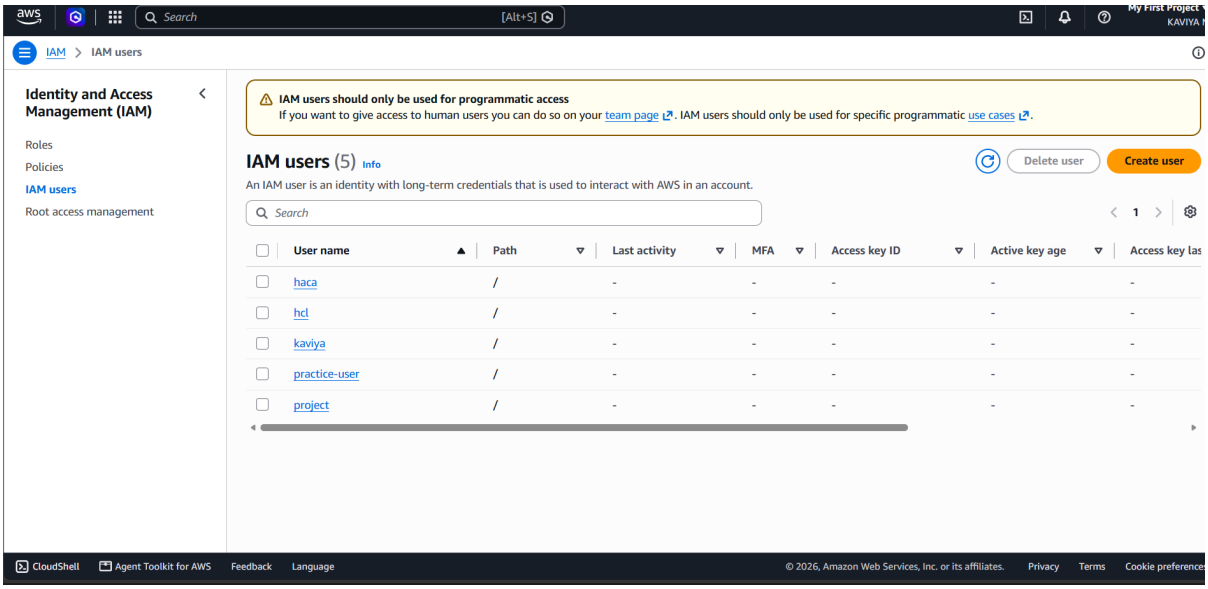
SECTION D: AWS CLOUD SERVICES

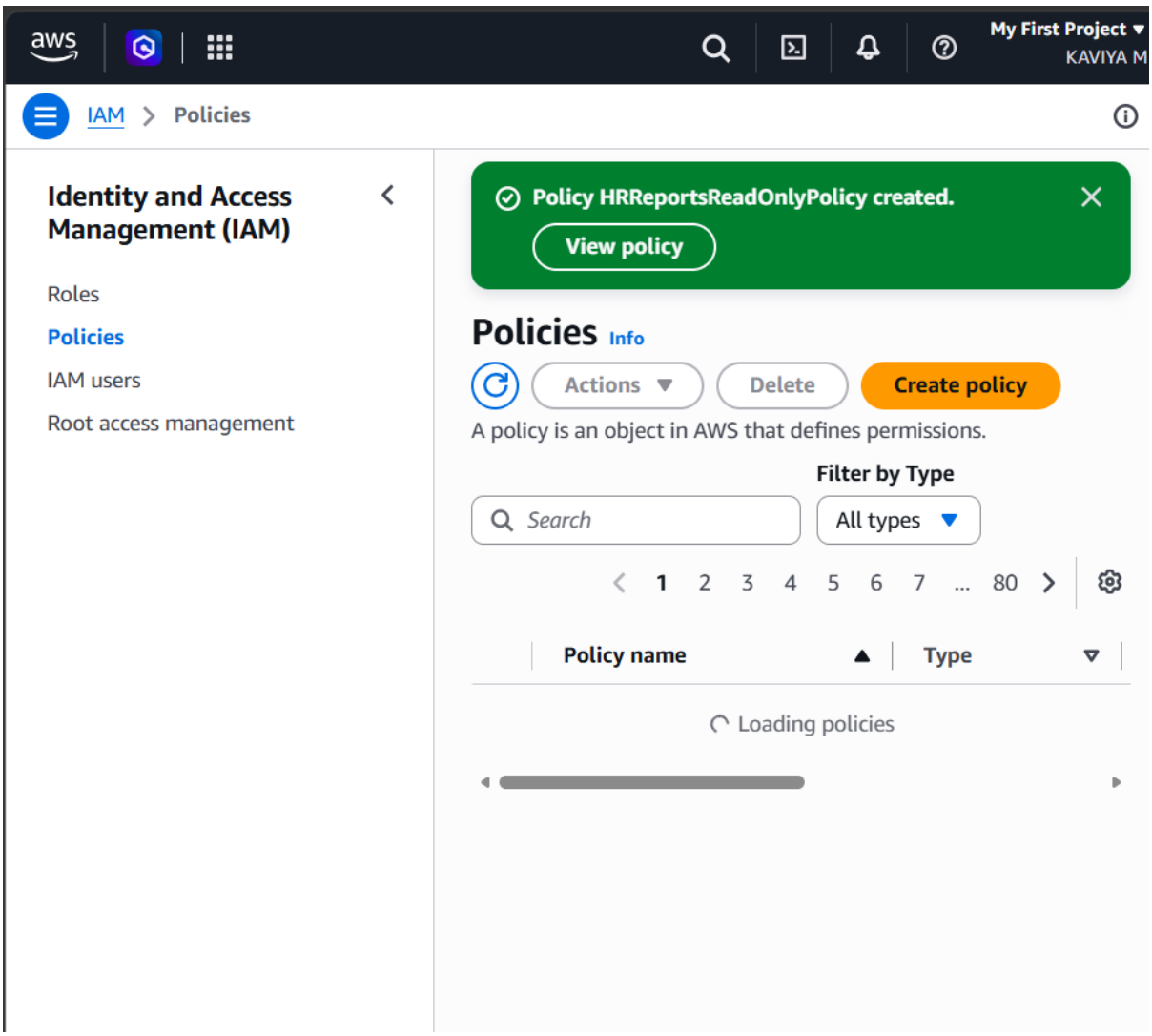
Question 10: S3 and IAM Security (5 Marks)

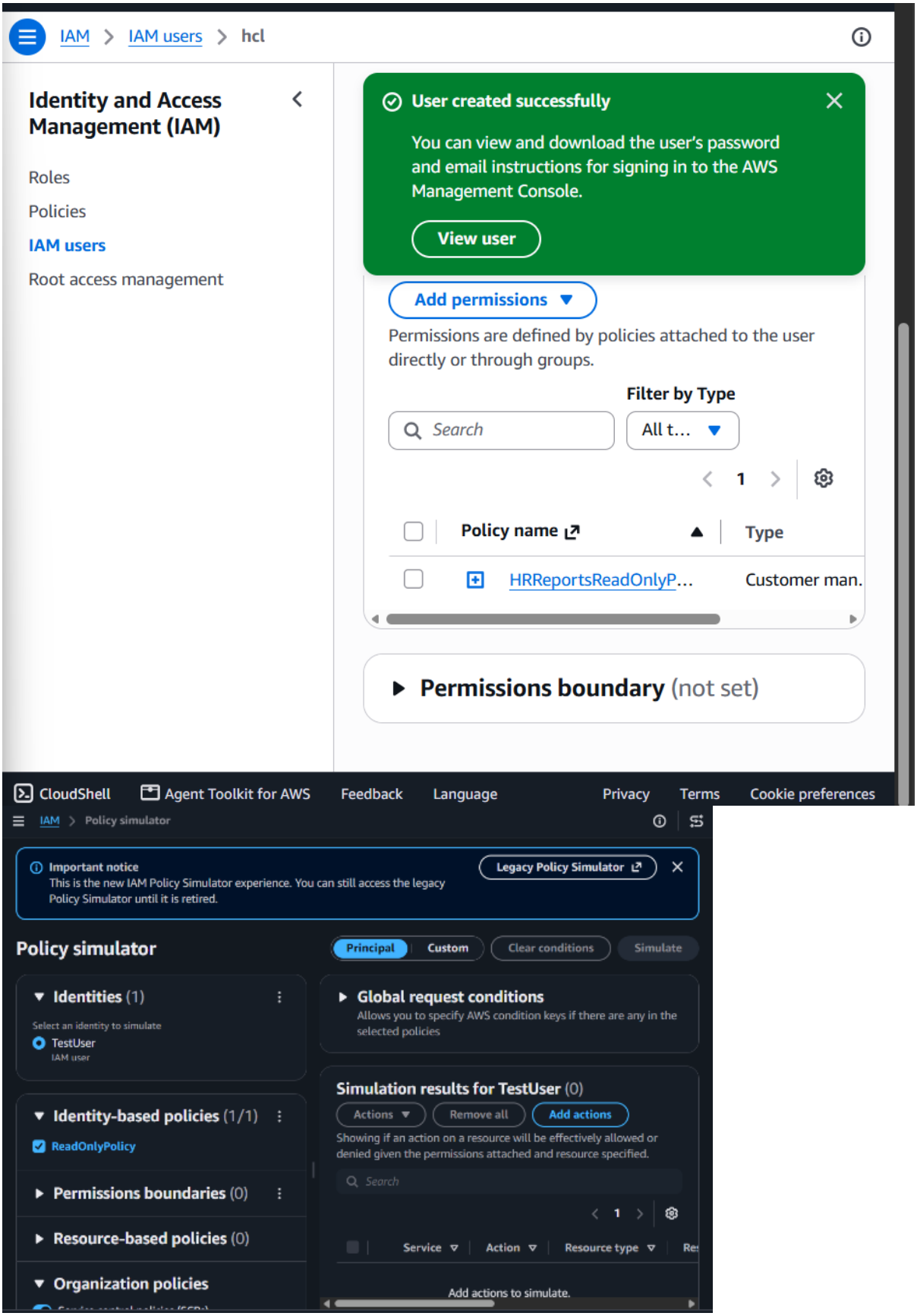
A company stores confidential employee reports in Amazon S3. Only authorized HR employees should be able to read these reports.


Search
[Alt+S]
My First Project
KAVIYA M
Amazon S3 > Buckets
Amazon S3
Buckets
Directory buckets
Table buckets
Vector buckets
Files
File systems
Access management and security
Access Points
Access Points for FSx
Access Grants
IAM Access Analyzer
Storage management and insights
Storage Lens
Batch Operations
Account and organization
Buckets
General purpose buckets
All AWS Regions
Directory buckets
General purpose buckets (11) Info
Copy ARN
Empty
Delete
Create bucket
Buckets are containers for data stored in S3.
Find buckets by name
Name
AWS Region
Creation date
amazon-sagemaker-250224372156-ap-southeast-2-bh2np7dn9rg1wi
Asia Pacific (Sydney) ap-southeast-2
October 8, 2026, 11:21:12 (UTC+05:30)
aws-glue-assets-250224372156-ap-southeast-2
Asia Pacific (Sydney) ap-southeast-2
October 11, 2026, 22:57:05 (UTC+05:30)
csvfiler
Asia Pacific (Sydney) ap-southeast-2
October 9, 2026, 09:58:49 (UTC+05:30)
hr-confidential-reports-2026-kav
Asia Pacific (Sydney) ap-southeast-2
October 28, 2026, 14:00:29 (UTC+05:30)
kaviyaproject
Asia Pacific (Sydney) ap-southeast-2
October 8, 2026, 00:20:43 (UTC+05:30)
kaviyas3-demo
Asia Pacific (Sydney) ap-southeast-2
October 8, 2026, 09:07:13 (UTC+05:30)
Account snapshot Info
Updated daily
View dashboard
Storage Lens provides visibility into storage usage and activity trends.
External access summary Info
Updated daily
External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.
CloudShell
Agent Toolkit for AWS
Feedback
Language
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


My First Project
KAVIYA M
Amazon S3
Upload succeeded
For more information, see the Files and folders table.
Summary
Destination
s3://hr-confidential-reports-2026-kav
Succeeded
0 files, 0 B (0%)
Failed
0 files, 0 B (0%)
Files and folders
Configuration
Files and folders (1 total, 97.0 B)
Find by name
1
Name
Folder
Type
Size
Status
Error
employee_re...
-
text/plain
97.0 B
Succeeded
-


Search
[Alt+S]
IAM
IAM users
Identity and Access Management (IAM)
Roles
Policies
IAM users
Root access management
IAM users should only be used for programmatic access
If you want to give access to human users you can do so on your team page. IAM users should only be used for specific programmatic use cases.
IAM users (5) Info
An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.
Delete user
Create user
Search
User name
Path
Last activity
MFA
Access key ID
Active key age
haca
hcl
kaviya
practice-user
project
CloudShell
Agent Toolkit for AWS
Feedback
Language
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


My First Project
KAVIYA M
IAM
Policies
Identity and Access Management (IAM)
Roles
Policies
IAM users
Root access management
Policy HRReportsReadOnlyPolicy created.
View policy
Policies Info
Actions
Delete
Create policy
A policy is an object in AWS that defines permissions.
Filter by Type
Search
All types
1 2 3 4 5 6 7 ... 80
Policy name
Type
Loading policies



c) Explain how you would verify that an unauthorized user cannot read the file. (1 mark)

I will sign in using an IAM test user who does not have permission to access the S3 bucket and try to download the file. AWS should display an Access Denied error, confirming that unauthorized users cannot read the file.

d) Explain how encryption at rest protects the stored data. (1 mark)

Encryption at rest protects stored data by converting it into unreadable ciphertext using encryption keys. Amazon S3 supports server-side encryption, such as SSE-S3. Only authorized requests can retrieve the data in its decrypted form, providing additional protection for confidential employee reports.

Question 11: EC2 Troubleshooting and CloudWatch (5 Marks)

a) Check the EC2 instance state and status checks. (1 mark)

Open the AWS Management Console and navigate to EC2 → Instances. Select the required instance and check whether its state is Running. Verify that both the system status check and instance status check are passing. If a check fails, investigate the underlying infrastructure or instance configuration.

b) Explain how security groups and route tables can affect application connectivity. (1 mark)

Security groups act as virtual firewalls that control inbound and outbound traffic. Incorrect rules may block application traffic on ports 80 or 443. Route tables determine the path of network traffic. Incorrect routes or a missing internet gateway route can prevent users from accessing the application.

c) Explain how to configure a CloudWatch alarm when CPU utilization exceeds 85% for a defined evaluation period. (2 marks)

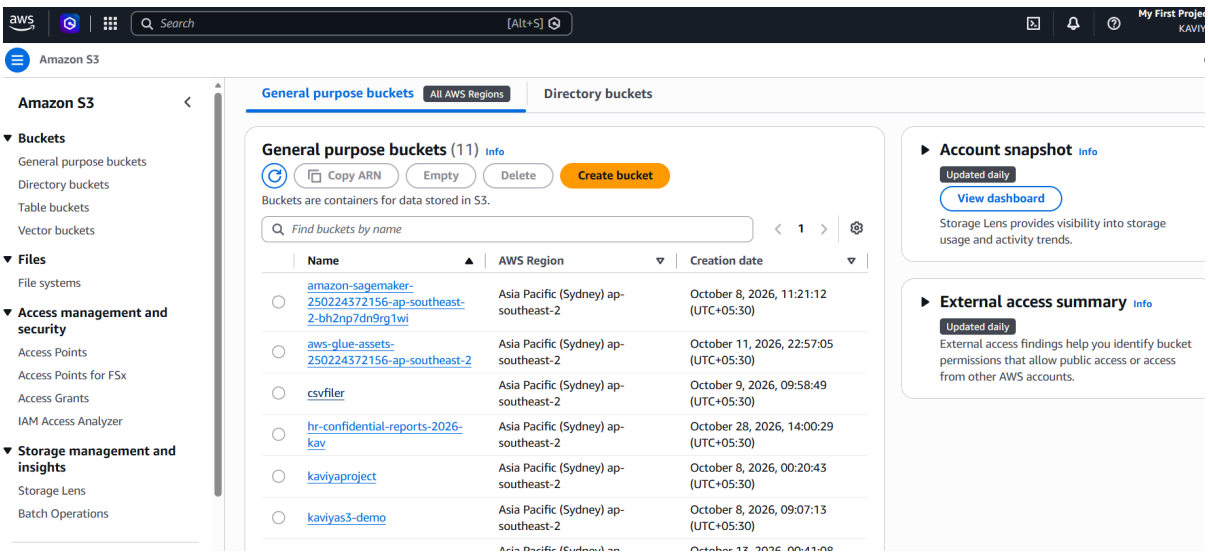
Open Amazon CloudWatch and select Alarms → Create alarm. Choose the EC2 CPUUtilization metric for the required instance. Set the threshold to greater than 85%, select an evaluation period, such as five minutes, and configure the required evaluation criteria. Optionally, configure an Amazon SNS notification. Finally, name and create the alarm. It enters the ALARM state when the configured threshold conditions are met.

d) Identify one method for investigating the root cause of high CPU utilization. (1 mark)

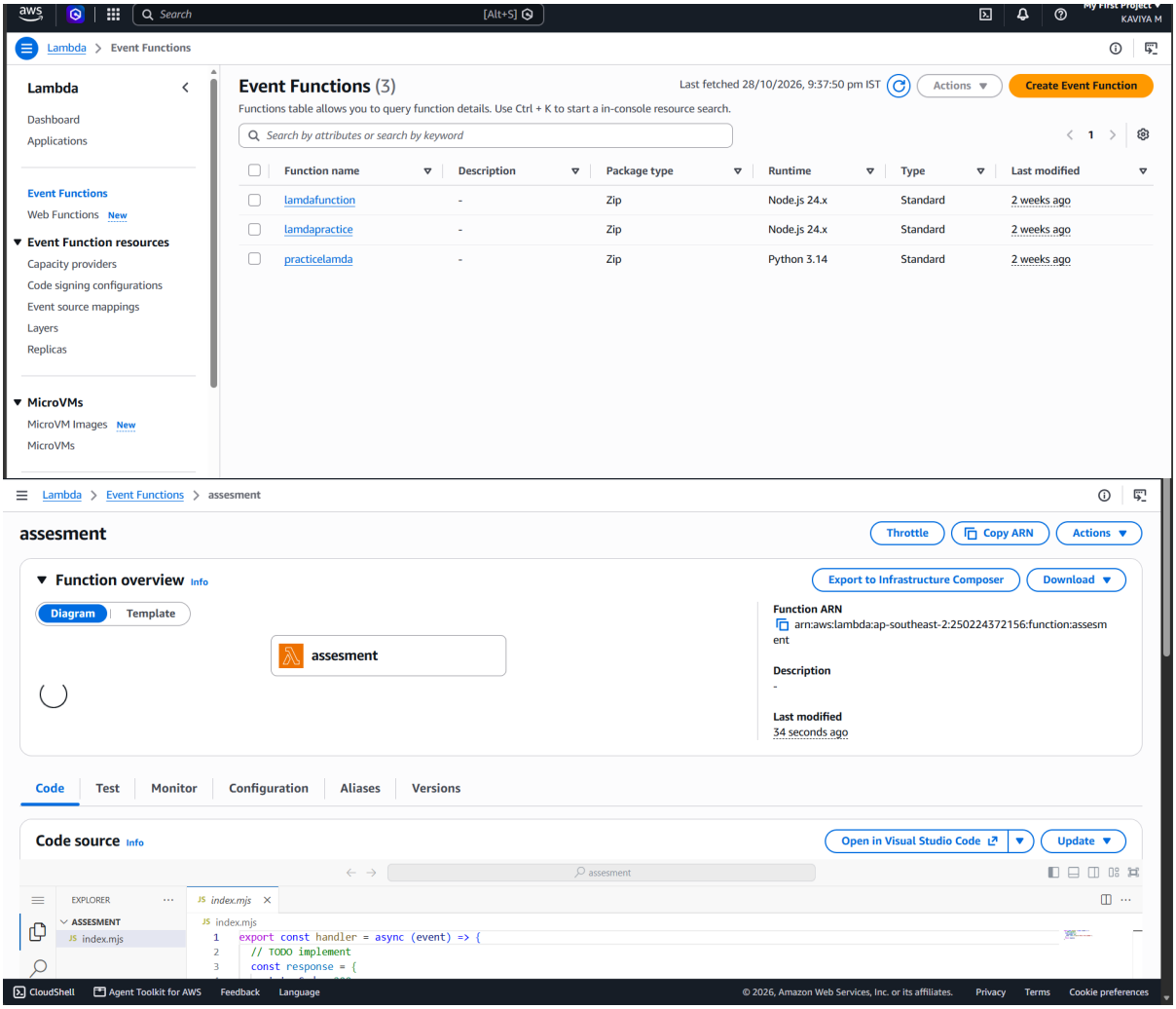
Use Amazon CloudWatch to monitor CPU utilization metrics over time. Analyze the periods of unusually high usage to identify when the issue occurs and help investigate the root cause of the performance problem.

Question 12: Event-Driven Processing (5 Marks)

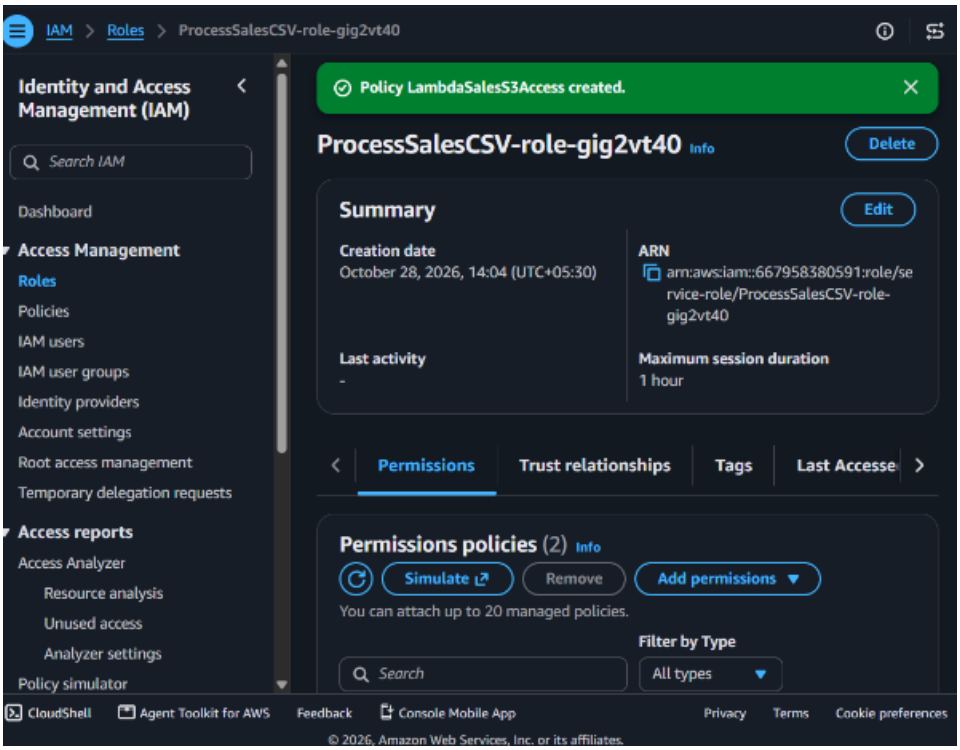
A company uploads CSV files to an S3 bucket. Each uploaded file must be validated and processed automatically, with the results stored in another location. Design a solution using AWS services.



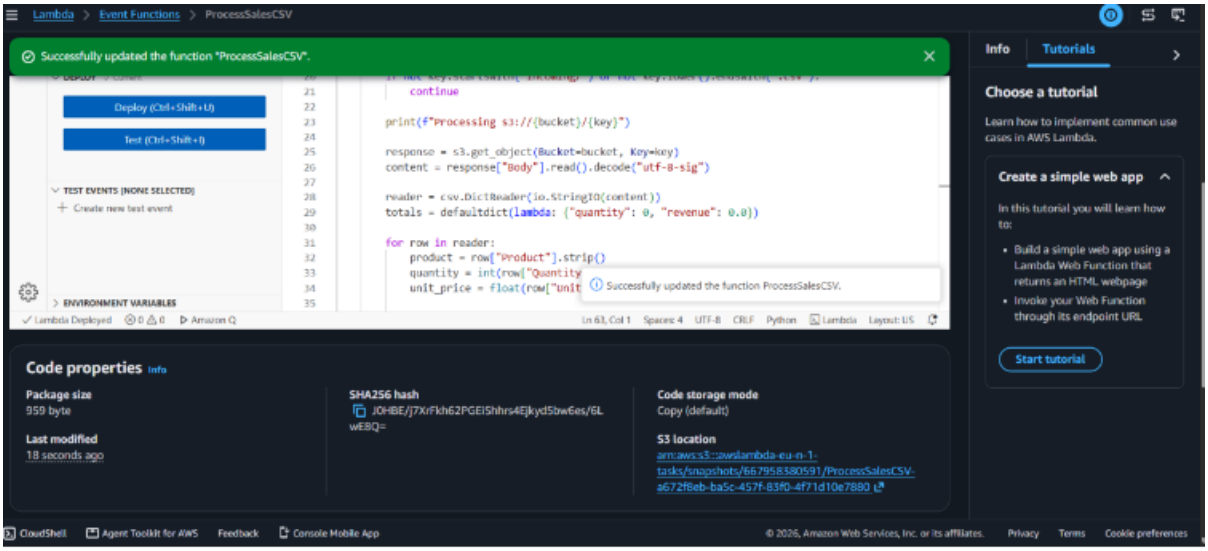
a) Identify a suitable service for executing the processing code. (1 mark)



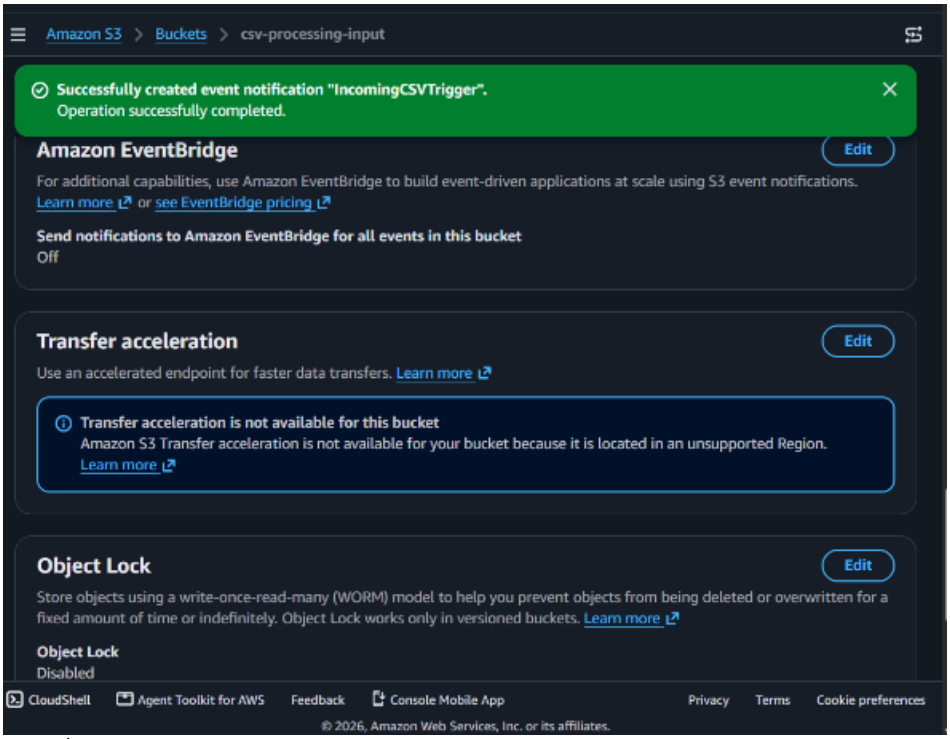
c) Explain the IAM permissions needed to read the input file and write the results. (1 mark)



a) Identify a suitable service for executing the processing code. (1 mark)



b) Explain how an S3 upload can trigger processing. (1 mark)



a) Identify a suitable service for executing the processing code. (1 mark)

b)

AWS Lambda is a suitable service for executing code to validate and process CSV files automatically without managing servers.

b) Explain how an S3 upload can trigger processing. (1 mark)

Configure an Amazon S3 Event Notification for the ObjectCreated event. Whenever a CSV file is uploaded to the input bucket, S3 automatically triggers the AWS Lambda function to process the file.

c) Explain the IAM permissions needed to read the input file and write the results. (1 mark)

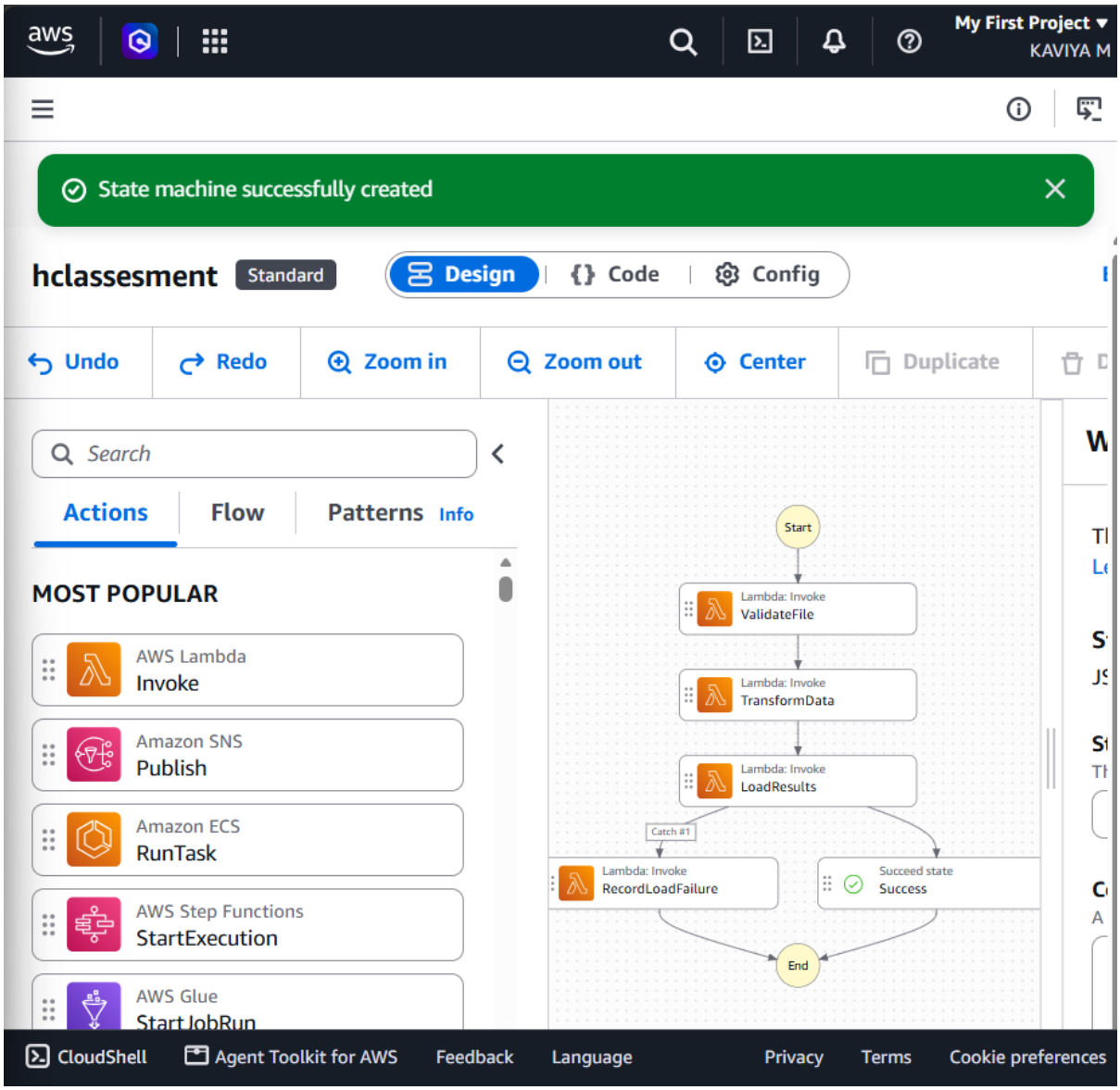
Create an IAM execution role for Lambda with permission to read input files using s3:GetObject on the input bucket and write processed results using s3:PutObject on the output bucket. Grant only the required permissions following the principle of least

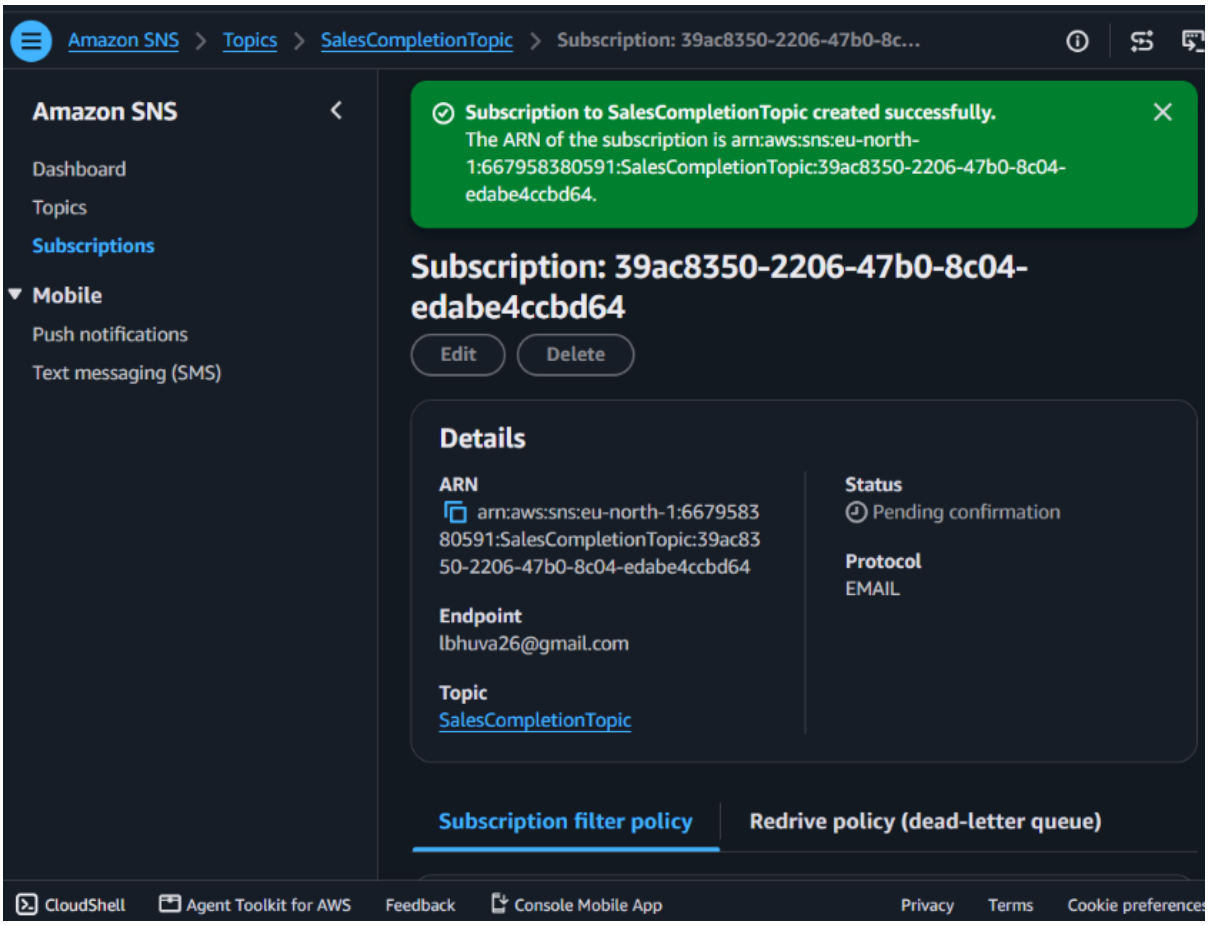
privilege.
d) Explain how errors and processing logs can be captured. (1 mark)

Use Amazon CloudWatch Logs to capture Lambda execution logs, errors, and processing details. CloudWatch can also monitor metrics and configure alarms to notify administrators when processing fails.
e) Explain how to prevent duplicate processing or recursive triggering. (1 mark)

Use idempotency checks to prevent processing the same file more than once. To prevent recursive triggering, store processed results in a separate output bucket or configure the S3 event notification to trigger only for CSV files in a specific input folder.
Question 13: AWS Step Functions — State Machine Design (7 Marks)

A company needs to process sales files through the following stages: 1. Validate the input file. 2. Transform the data. 3. Calculate total sales by product. 4. Store the results in Amazon S3. 5. Send a completion notification. Design an AWS Step Functions state machine to coordinate the workflow.
a. Identify suitable states for validation, processing, storage, and notification. (2 marks)
b. Explain how a Choice state can route invalid files to a failure path and valid files to processing. (1 mark)
c. Explain how Retry and Catch can handle transient failures and permanent errors. (2 marks)
d. Explain how the state machine can invoke Lambda functions or other supported AWS services. (1 mark)
e. Explain how execution history helps troubleshoot a failed workflow. (1 mark)


My First Project
KAVIYA M
State machine successfully created
hclassesment
Standard
Design
Code
Config
Undo
Redo
Zoom in
Zoom out
Center
Duplicate
Search
Actions
Flow
Patterns
Info
MOST POPULAR
AWS Lambda
Invoke
Amazon SNS
Publish
Amazon ECS
RunTask
AWS Step Functions
StartExecution
AWS Glue
Start JobRun
Start
Lambda: Invoke
ValidateFile
Lambda: Invoke
TransformData
Lambda: Invoke
LoadResults
Catch #1
Lambda: Invoke
RecordLoadFailure
Succeed state
Success
End
CloudShell
Agent Toolkit for AWS
Feedback
Language
Privacy
Terms
Cookie preferences



Verify the State Machine

Open the AWS Step Functions console and select the created state machine. Click Start execution and enter the input JSON containing the S3 bucket name and CSV file path, such as incoming/sales.csv. Replace the sample bucket name with your actual S3 bucket name and click Start execution. Monitor the execution graph to ensure that the ValidateInput, CheckValidation, TransformData, CalculateSales, StoreResults, and SendNotification states complete successfully. Finally, open the S3 bucket's processed/ folder and verify that sales_summary.csv has been created. Check your email for the SNS completion notification to confirm that the workflow has finished successfully.

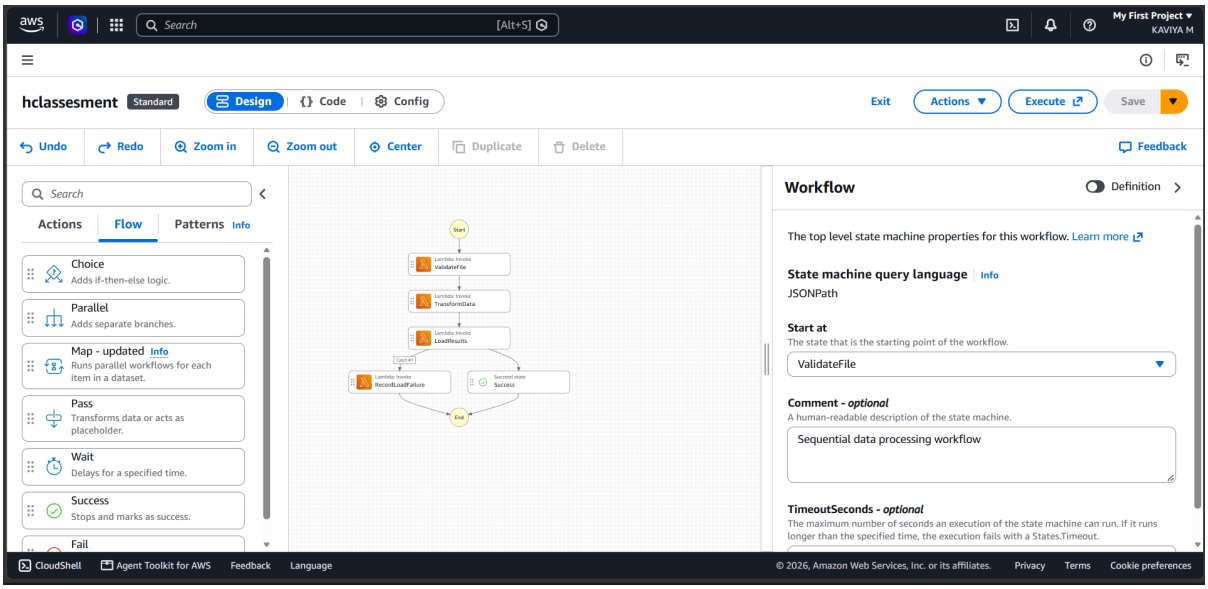
Question 14: Step Functions Implementation

A data engineering team has developed three processing functions: ValidateFile, TransformData and LoadResults. The functions must execute in sequence. If validation fails, the workflow must stop. If transformation fails temporarily, the workflow must retry. If loading fails permanently, the workflow must record the failure.

a. Write an ASL JSON definition containing the three sequential tasks. Use placeholders for resource ARNs where necessary. (3 marks)

b. Configure a retry policy for the transformation task with an appropriate error category, retry interval, and maximum attempts. (2 marks)

c. Configure a Catch transition for the loading task that routes to an error-handling state. (1 mark)



In AWS Step Functions, data is passed between tasks as JSON input and output. The output of one task becomes the input of the next task. The ResultPath field stores a task's result in the execution data, while Payload.$ passes the current JSON data to a Lambda function. This allows each task to use the results produced by previous tasks during workflow execution.

e. Explain how you would test successful and failed executions.

To test successful execution, start the state machine with valid input data and verify that ValidateFile, TransformData, and LoadResults execute in sequence and finish successfully. To test failed execution, provide invalid input to make validation fail, simulate a temporary transformation error to verify retries, and simulate a loading error to verify that the Catch transition records the failure. Check the AWS Step Functions execution history to inspect state transitions, retry attempts, and error details.

SECTION F: MICROSOFT AZURE
Question 15: Azure Data Lake Storage and Security

Step 1: Open Azure Portal

Open the Microsoft Azure Portal and sign in using your Azure account. Search for Storage accounts in the top search bar and select the service to begin creating a storage account.

Step 2: Create a Storage Account

Click Create and select your subscription and resource group. Enter a globally unique storage account name, choose a region, select Standard performance, and configure the required redundancy option.

Step 3: Enable Hierarchical Namespace

In the Advanced tab, find the Hierarchical namespace option and enable it. This feature enables Azure Data Lake Storage Gen2 capabilities, including directory-based organization and file-system semantics. Click Review + create, validate the configuration, and click Create.

Step 4: Create a Container

After deployment, open the storage account and navigate to Data storage → Containers. Click + Container, enter the name salesdata, and set the anonymous access level to Private (no anonymous access). Click Create to create the container securely.

Step 5: Create a Sample Dataset

Open Notepad on your computer and enter sample employee data with columns such as EmployeeID, EmployeeName, Department, and Salary. Save the file as employees.csv, selecting All files in the Save As dialog if necessary.

Step 6: Upload the Dataset

Open the salesdata container in Azure Portal and click Upload. Browse your computer, select employees.csv, and click Upload. Verify that the uploaded file appears in the container.

Step 7: Configure Access Using Microsoft Entra ID and Azure RBAC

Open the storage account and select Access control (IAM). Click Add → Add role assignment, select the Storage Blob Data Reader role, and choose the authorized user or group from Microsoft Entra ID. Click Review + assign to apply the permissions.

Step 8: Verify Authorized Access

Sign in as the authorized user or use an authenticated storage client. Open the salesdata container and attempt to read or download employees.csv. Confirm that the user can access the dataset with the assigned read-only permissions.

Step 9: Verify Unauthorized Access

Use a separate test identity that has no applicable data-access permissions. Attempt to access the same dataset and confirm that Azure denies access. Keep the container private and check for any inherited roles or other permissions that might grant access.

Question 16: Azure VM Troubleshooting

Step 1: Verify VM Provisioning and Power State

Open the Azure Portal and navigate to Virtual Machines. Select the required VM and check its provisioning status and power state. Ensure that the VM is successfully provisioned and running.

Step 2: Check Network Security Group (NSG) Rules

Open the VM's Networking settings and review the associated NSG rules. Check whether the required inbound and outbound ports are allowed. Incorrect rules may block connectivity to the VM.

Step 3: Verify Subnet and Routing Configuration

Check the virtual network and subnet assigned to the VM. Review the route table and verify that traffic is directed to the correct destination. Incorrect subnet or routing configurations can prevent network communication.

Step 4: Use Azure Network Troubleshooting Tools

Use Azure Network Watcher to diagnose connectivity issues and inspect effective security rules. Identify blocked traffic and correct the relevant network configuration. Test the connection again after making changes.

Step 5: Follow Secure Troubleshooting Practices

Avoid exposing VM management ports directly to the public internet. Use Azure Bastion for secure SSH or RDP access and restrict network access to trusted sources. Apply the principle of least privilege.

Question 17: Azure Data Integration

Step 1: Create Azure Data Factory

Open the Azure Portal and search for Data Factory. Create a Data Factory instance by selecting the subscription, resource group, region, and name. Open Azure Data Factory Studio to create and manage pipelines.

Step 2: Connect CSV Files to ADLS Gen2

Create a linked service to connect Azure Data Factory to Azure Data Lake Storage Gen2. Configure the source dataset to read daily customer transaction CSV files. Set the destination dataset to the required storage container.

Step 3: Organize Data into Date-Based Folders

Configure the pipeline to store files in date-based folders, such as year=2026/ month=10/day=28. This structure helps organize daily transaction data. It also makes searching and processing data for specific dates easier.

Step 4: Handle Malformed and Missing Records

Validate CSV records for missing values, incorrect data types, invalid dates, and missing required fields. Separate invalid records from valid records during processing. Store rejected records in a separate quarantine folder for further investigation.

Step 5: Monitor Pipeline Execution

Open the Monitor section in Azure Data Factory Studio to check pipeline status. Review failed activities, error messages, and execution details. Configure alerts for failures so that issues can be identified and resolved quickly.

Step 6: Validate the Output Data

Check the processed files in ADLS Gen2 to confirm that records are stored correctly. Compare the source, successfully processed, and rejected record counts. Verify that valid data is stored in the correct folders and invalid records are separated.